Organofluorine Compounds

Modern Synthetic Methods for Fluorine-Substituted Target Molecules

*Masaki Shimizu and Tamejiro Hiyama**

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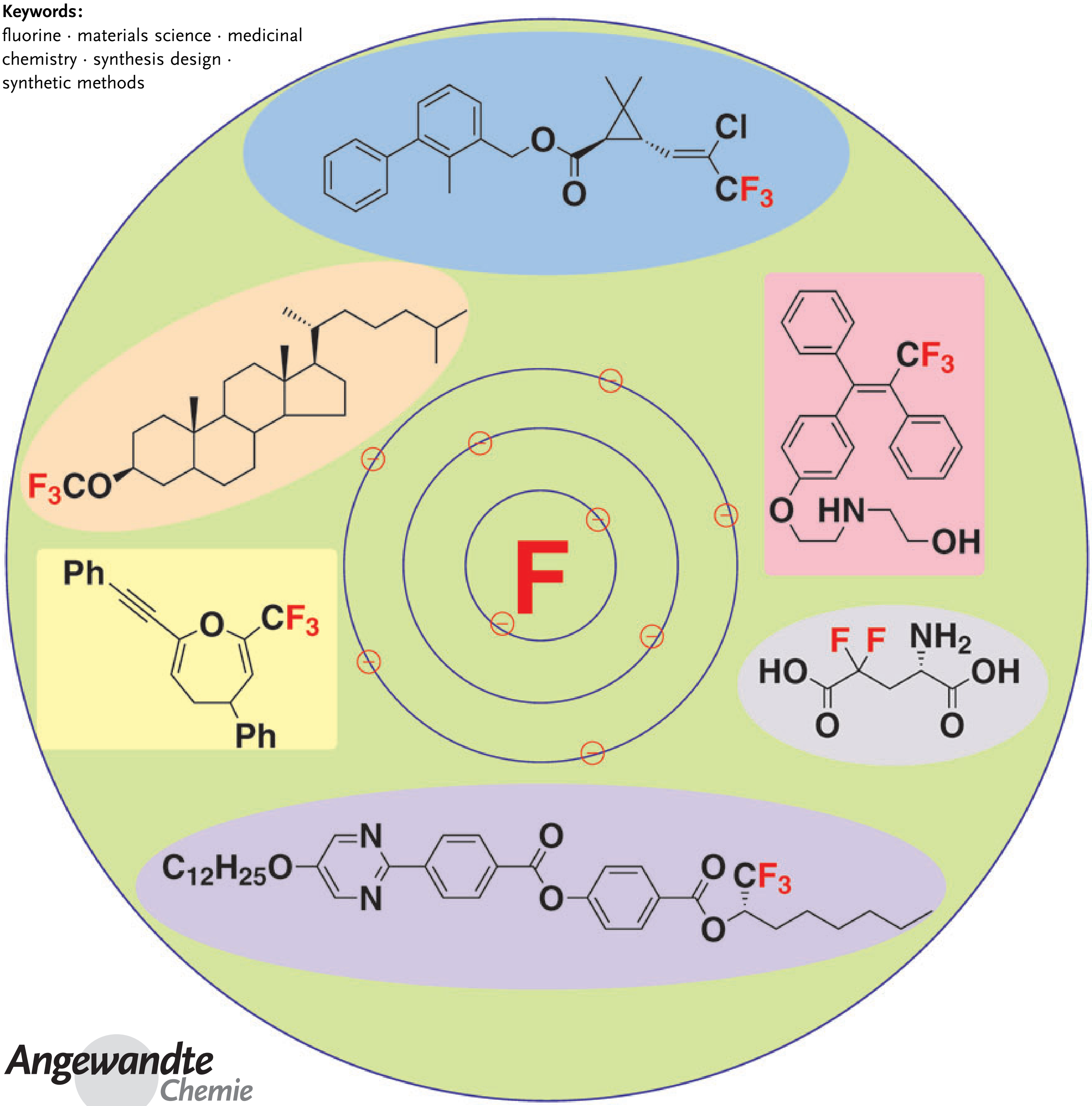

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Fluorine has come to be recognized as a key element in materials science: in heat-transfer agents, liquid crystals, dyes, surfactants, plastics, elastomers, membranes, and other materials. Furthermore, many fluorine-containing biologically active agents are finding applications as pharmaceuticals and agrochemicals. Progress in synthetic fluorine chemistry has been critical to the development of these fields and has led to the invention of many novel fluorinated molecules as future drugs and materials. As a result of the electronic effects of fluorine substituents, fluorinated substrates and reagents often exhibit unusual and unique chemical properties, which often make them incompatible with established synthetic methods. Thus, the problem of how to control the unusual properties of compounds with fluorine substituents deserves much attention, so as to promote the design of facile, efficient, and environmentally benign methods for the synthesis of valuable organofluorine targets.

From the Contents

1. Introduction

As the most electronegative element, fluorine has played a key role in recent pharmaceutical, agrochemical, and materials science. As the incorporation of fluorine and/or fluorine-containing groups into an organic molecule often drastically perturbs the chemical, physical, and biological properties of the parent compound (Figure 1), it is thought that the synthesis of desired fluorinated compounds should lead to the invention of novel agents and materials endowed with prerequisite functions.[1] Since a basic discussion of such fluorine effects is found in many books and reviews,[1,2] we do not cover them in detail herein. Instead we draw your attention to the very recent special issue of *ChemBioChem* on "Fluorine in the Life Sciences", which provides a very comprehensive and up-to-date picture of fluoroorganic compounds and fluorine effects.[3]

Because naturally occurring organofluorine compounds are rare, target organofluorine compounds are accessible solely by organic synthesis. With respect to chemical reactivity and stability, fluorinated compounds behave in a unique manner owing to so-called inductive and resonance effects caused by fluorine.[4] For example, fluorine-substituted carbocations are stabilized by the donation of nonbonding electron pairs on the fluorine atom, whereas electrostatic p–n repulsion between an anionic center and the lone pairs of electrons on the fluorine atom destabilizes a fluorine-substituted anionic center (Figure 2). Similarly, n–π repulsion between the lone pairs and a π-electron system pushes the π electrons far apart. Furthermore, a resonance effect operates in which the lone pairs participate in conjugation with the π system.

Like various other fluorine-containing groups, trifluoromethyl groups at C(sp^2) and C(sp^3) centers always withdraw electrons to destabilize cations and to stabilize anions (Figure 3). This anion stabilization can also be understood to be a result of negative hyperconjugation, an overlap of an anionic p orbital with the σ^* orbital of a C–F bond.[5]

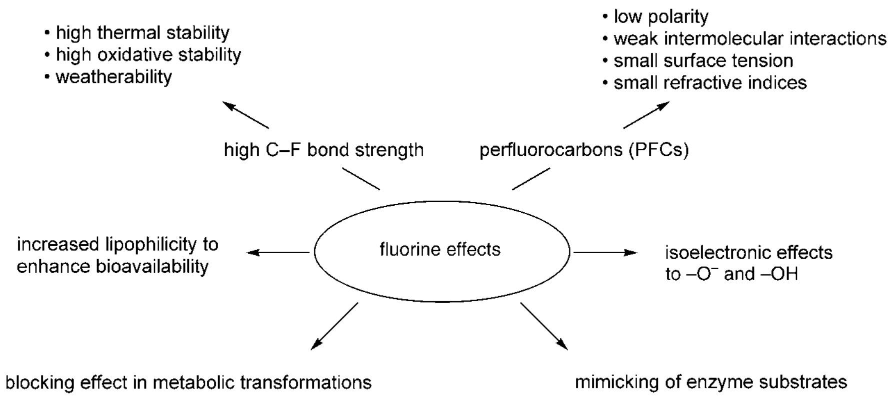

Figure 1. Fluorine effects in biologically active agents and materials.

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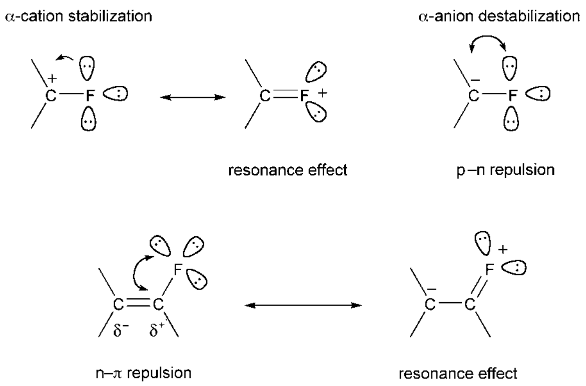

Figure 2. α-Fluorine effects.

Figure 3. α-CF_3 effects on carbocations and carbanions.

Trifluoromethyl groups also strongly affect chemical reactivity. For example, nucleophilic substitution at a trifluoromethyl-substituted carbon atom is not an easy process. The difficulty is attributed to destabilization of the transition state by fluorine and also to electrostatic repulsion between a nucleophile and the lone pairs of electrons on the fluorine atoms. There are many phenomena that can be understood in terms of C–H···F or C–M···F interactions, although such interactions are still a topic of controversy.[6]

In view of the unique characteristics of fluorine, it is easy to recognize that conventional synthetic methods are not always applicable to the synthesis of organofluorine compounds. Thus, a deep understanding of the nature of reaction mechanisms, reagents, and synthetic strategies based on the unique chemical and physical properties of fluorine is essential.[7] Herein we describe some novel synthetic methods, particularly those we have developed over the last two decades for the preparation of fluorine-substituted target molecules, focusing on oxidative desulfurization–fluorination, synthetic reactions based on fluorine-containing polyhaloalkyl metal reagents and CF_3-substituted oxiranyl lithium reagents, and the asymmetric synthesis of 1,1,1-trifluoro-2-alkanols.

2. Classification of Synthetic Methods

Synthetic methods for the synthesis of organofluorine compounds are roughly classified into two basic strategies: the fluorination method and the building-block method. Fluorination includes C–F bond formation through the transformation of a C–H bond or a functional group by treatment with a fluorinating reagent, such as elemental fluorine (F_2), hydrogen fluoride (HF), or derivatives thereof.[8] This is a straightforward method for the introduction of fluorine atoms into target molecules, especially when the fluorination can be carried out with high regio- and chemoselectivities. However, the handling of such fluorinating reagents requires special apparatus, techniques, and care, as the reagents are usually extraordinarily reactive and toxic, often as a result of corrosive HF that is released. Hence, methods that employ reagents with moderate reactivity under mild conditions are needed for the facile synthesis of fluorine-containing target molecules.

The building-block method involves appropriate chemical transformations starting from fluorinated small molecules (building blocks).[9] Since standard experimental procedures are applicable, this approach is more familiar to synthetic chemists. The ready availability and easy handling of fluorine-containing building blocks are essential for the success of this approach. However, fluorine substitution very often causes abnormal behavior with regard to the reactivity and/or selectivity of these building blocks relative to that of their nonfluorinated counterparts, thus providing a challenge to synthetic chemists to overcome these fluorine effects.

3. Oxidative Desulfurization–Fluorination

Many reagents for nucleophilic fluorination are now available. Representative examples are hydrogen fluoride (HF, b.p. 19.54 °C), its aqueous solution (hydrofluoric acid), sulfur tetrafluoride (SF_4), and diethylaminosulfur trifluoride (DAST); these reagents are, however, very reactive, corro-

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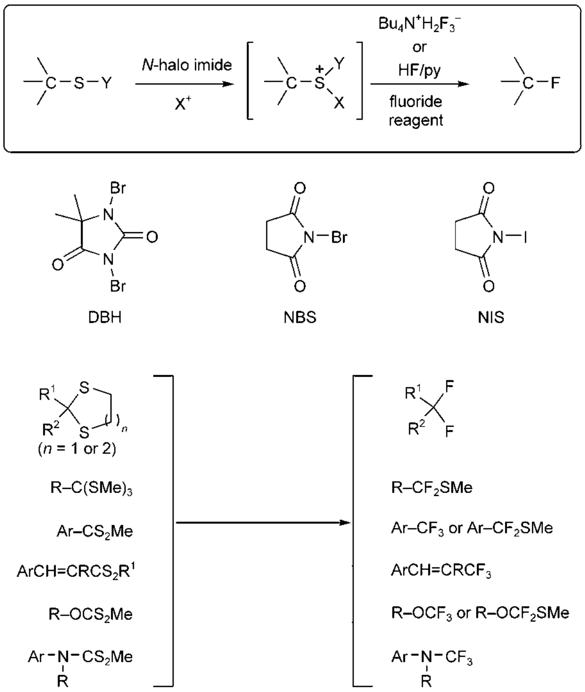
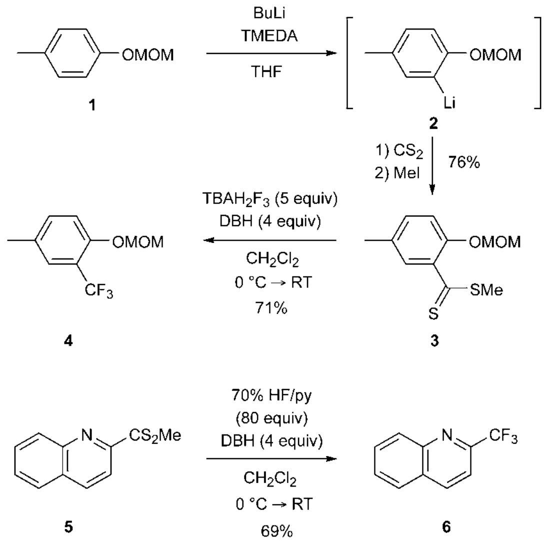
Masaki Shimizu was born in 1965 in Tokyo. He completed his PhD at Tokyo Institute of Technology in 1994 (Takeshi Nakai and Koichi Mikami). He was a research associate at Mitsubishi Chemical Co., then from 1995 at Tokyo Institute of Technology. In 1998, he moved to Kyoto University as Assistant Professor. He spent one year at Massachusetts Institute of Technology as a postdoctoral fellow (S. L. Buchwald) and was promoted to Associate Professor at Kyoto University in 2003. His research interests focus on synthetic methods with organofluorine compounds and organometallic reagents, and functional organosilicon materials.

sive, and toxic. When they are modified for more convenient handling, their nucleophilicity often decreases. To facilitate nucleophilic attack of a substrate by the modified reagent, electrophilic substrate activation by an oxidant is quite effective, as exemplified in the halofluorination of alkenes and alkynes,[10] the thio- and selenofluorination of alkenes,[11] and nitrofluorination.[12] We have applied this concept to organosulfur compounds, a readily accessible class of organic compounds (Scheme 1).[13] The basic idea involves activation of a C–S bond by a positive halogen generated from N-iodosuccinimide (NIS), N-bromosuccinimide (NBS), or 1,3-dibromo-5,5-dimethylhydantoin (DBH), followed by nucleophilic substitution with a fluoride ion from HF/pyridine (Olah reagent) or tetrabutylammonium dihydrogen trifluoride ($TBAH_2F_3$). This methodology is called oxidative desulfurization–fluorination and is applicable to the preparation of various types of organofluorine compounds. The scope is summarized in Scheme 1. The following paragraphs present some representative examples that demonstrate the versatility of this method.



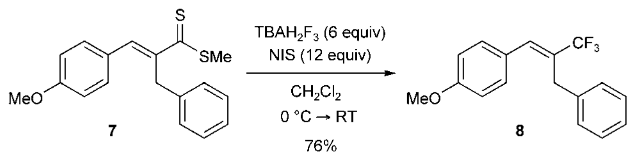
Scheme 1. Oxidative desulfurization–fluorination. py = pyridine.

Trifluoromethyl-substituted aromatic compounds can be prepared from the corresponding arenedithiocarboxylates with $TBAH_2F_3$ and DBH.[14] For example, the *ortho* lithiation of *p*-cresol methoxymethyl (MOM) ether (**1**), followed by trapping of the resulting anion **2** with carbon disulfide and then with iodomethane, produces the dithiocarboxylate **3**, which is smoothly converted into the trifluoromethyl-substituted *p*-cresol derivative **4** in good overall yield (Scheme 2). The trifluorination of dithiocarboxylates is applicable to heteroaromatic compounds, such as the quinoline derivative **5**: A combination of DBH and 70 % HF/pyridine as the fluoride source was the best system for the transformation of electron-deficient heteroaromatic compounds.



Scheme 2. Synthesis of trifluoromethyl-substituted aromatic compounds. TMEDA = *N,N,N',N'*-tetramethylethylenediamine.

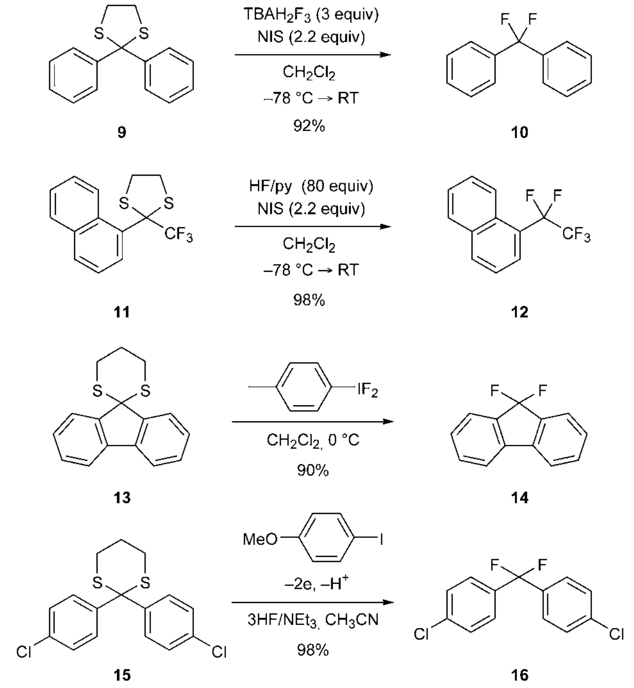
The α,β-unsaturated dithiocarboxylate **7** undergoes trifluoromethylation with a combination of $TBAH_2F_3$ and NIS, giving rise to the 1-aryl 3,3,3-trifluoropropene **8** (Scheme 3).[14] Other combinations (HF/pyridine in place of $TBAH_2F_3$; DBH or NBS in lieu of NIS) were ineffective for this transformation.



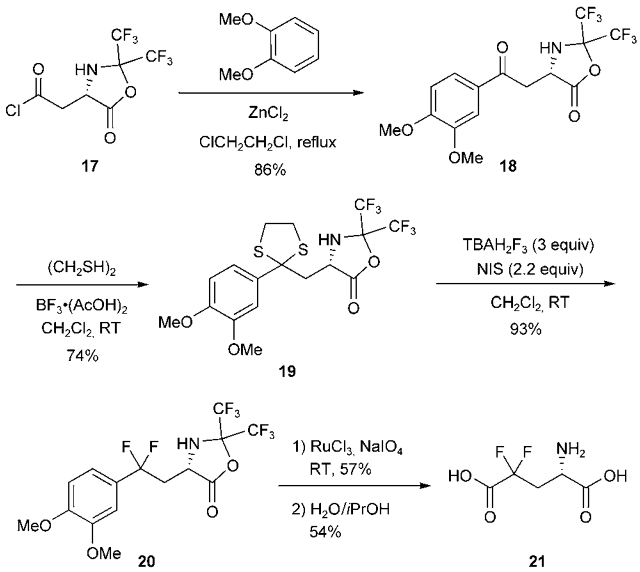
Scheme 3. Synthesis of the 1-aryl 3,3,3-trifluoropropene **8**.

Since the introduction of a *gem*-difluoromethylene moiety into biologically active but unstable agents considerably improves their biostability, there is a need for facile and versatile methods for the construction of difluoromethylene groups.[15] Upon treatment with NIS and $TBAH_2F_3$ or HF/pyridine, the dithioacetals of benzophenone **9** and 1-naphthyl trifluoromethyl ketone **11** afford **10** and **12**, respectively, in high yields (Scheme 4).[16] When DBH is employed as the oxidant, the difluorination is sometimes accompanied by halogenation of the aromatic ring. The same type of transformation can be performed with p-$CH_3C_6H_4IF_2$ and electrochemically generated p-$(MeO)C_6H_4IF_2$ (Scheme 4).[17] The synthetic utility of this strategy is demonstrated by the synthesis of the difluoroglutamic acid **21** (Scheme 5).[18]

Difluoroalkyl ethers and difluoromethylenedioxy compounds can be prepared from thione esters and thione carbonates, which are in turn readily accessible from esters and carbonates by treatment with the Lawesson reagent ((p-$(MeO)C_6H_4P(=S)S)_2$).[19] For example, in the presence of



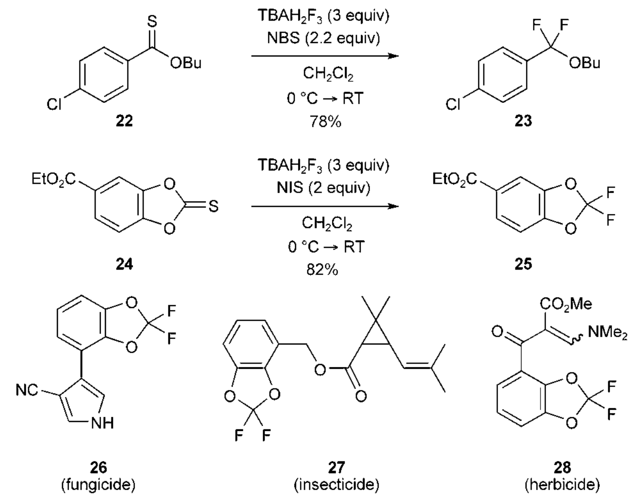
Scheme 4. Oxidative desulfurization–fluorination of dithioacetals derived from aryl ketones.



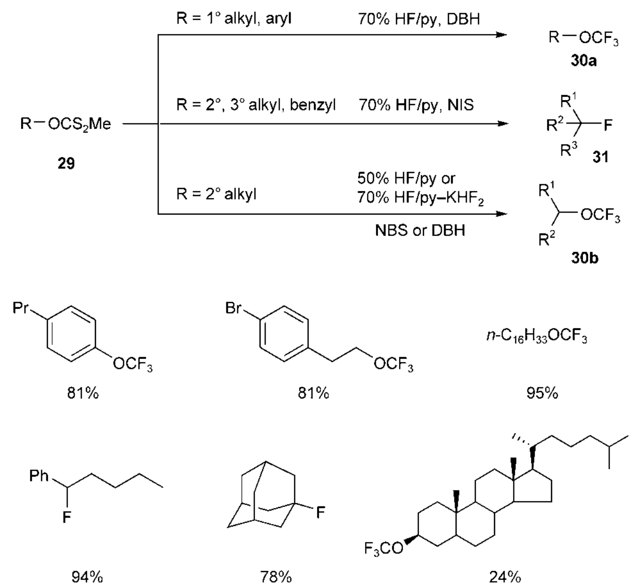
Scheme 5. Synthesis of the difluoroglutamic acid **21**.

$TBAH_2F_3$ and NBS (or NIS) the thione ester **22** and carbonate **24** are converted into **23** and **25**, respectively, in good yields (Scheme 6). This approach is useful for the preparation of biologically active substances containing a difluoromethylenedioxy moiety, such as **26**, **27**, and **28**.

Trifluoromethyl ethers **30a** of primary alcohols and phenols can be prepared from the corresponding xanthates **29** with 70% HF/pyridine and DBH (Scheme 7).[20] In contrast, the treatment of xanthates derived from secondary or tertiary alcohols with 70% HF/pyridine and NIS results in



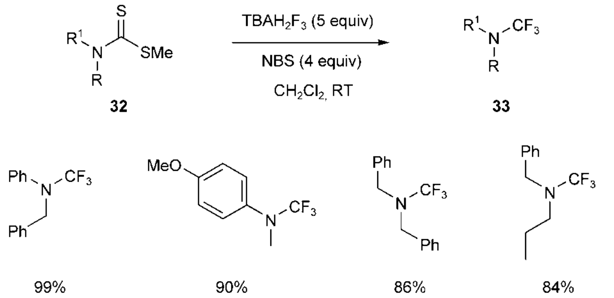
Scheme 6. Preparation of difluoroalkyl ethers and difluoromethylenedioxy compounds.



Scheme 7. Synthesis of trifluoromethyl ethers and alkyl fluorides from dithiocarbonates.

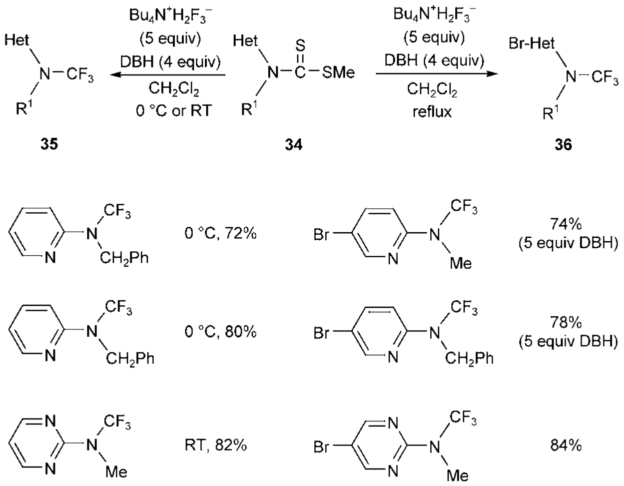
fluorination, giving rise to secondary or tertiary fluorides **31**, respectively. However, the use of milder fluorinating reagents, such as 50% HF/pyridine or 70% HF/pyridine with KHF_2, for the reaction of secondary xanthates **29** allows the isolation of secondary alkyl trifluoromethyl ethers **30b**. This reaction is in sharp contrast with that carried out in the presence of $TolIF_2$ in which only fluorination products are formed.[21] 3β-Trifluoromethoxycholestane prepared by this method was found to act as a chiral dopant for twisted-nematic liquid-crystal displays.[22] Cyclohexyl trifluoromethyl ethers prepared in a similar way are potential future materials for TFT (thin-film transistor for active matrix) displays.[23]

Dithiocarbamates **32**, readily available from secondary amines, carbon disulfide, and methyl iodide, are converted into trifluoromethyl amines **33** by treatment with $TBAH_2F_3$ and NBS (Scheme 8).[23] This protocol is applicable to both aromatic and aliphatic amines. Aryl trifluoromethylamines are stable to moisture, whereas dialkyl trifluoromethylamines are easily hydrolyzed. The presence of a trifluoromethyl group on the nitrogen atom retards oxidation of the amino group.



Scheme 8. Synthesis of trifluoromethylamines.

The trifluoromethylation of heteroarenedithiocarbamates **34** with DBH at 0 °C or room temperature gives trifluoromethylamines **35**, whereas the same reaction in CH_2Cl_2 at reflux causes selective bromination of the aromatic ring at the *para* position as well as trifluorination to produce **36** (Scheme 9).[23]



Scheme 9. Oxidative desulfurization–fluorination of heteroaryl dithiocarbamates.

4. Fluorine-Containing Polyhaloalkyl Metal Reagents

Since halogen–metal exchange is a relatively facile process, the reduction of commercially available fluorine-containing polyhaloalkanes, so-called halons, with a metal or an organometallic reagent would be a versatile method for the generation of fluorine-containing carbenoids, if the stability and reactivity of the reagents could be controlled.[24]

The artificial pyrethroids bifenthrin and cyhalothrin, both of which contain $CH{=}C(Cl)CF_3$ in lieu of $CH{=}CMe_2$ or $CH{=}CCl_2$, are highly potent, exhibiting remarkably enhanced activity relative to nonfluorinated congeners (Figure 4).[25] We have established a practical and stereocontrolled approach to

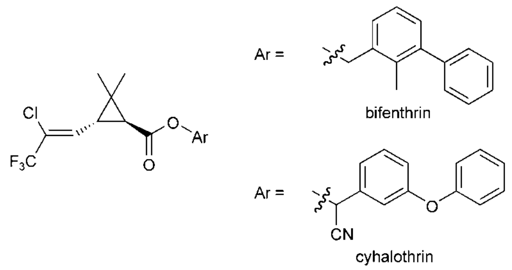
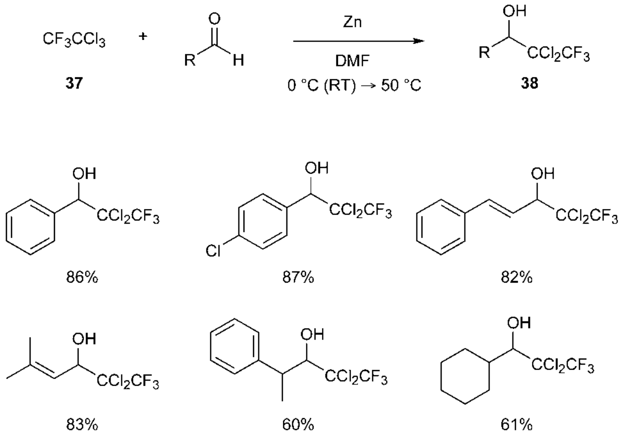


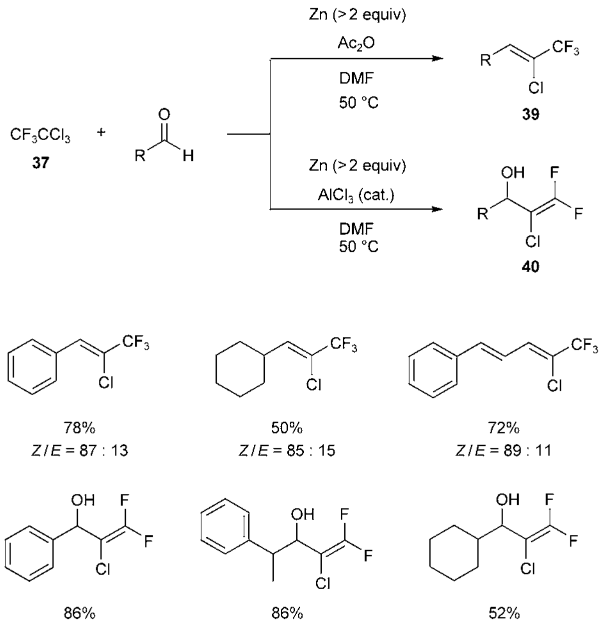
Figure 4. Examples of artificial pyrethroids.

di- and trifluorinated synthetic pyrethroids based on the addition of CF_3CCl_2ZnCl to aldehydes and reductive olefination of the adducts.[26] The two-step transformation can also be carried out in one pot. The reduction of CF_3CCl_3 (**37**) with zinc metal in DMF at 50 °C generates the thermally stable zinc carbenoid CF_3CCl_2ZnCl, which reacts with various aldehydes, including aromatic, aliphatic, and α,β-unsaturated aldehydes, to give alcohols **38** in good yields (Scheme 10). A similar transformation was disclosed independently by Lang and coworkers, who were successful in isolating the carbenoid and showed it to be monomeric in the solid state by single-crystal X-ray diffraction.[27,28]



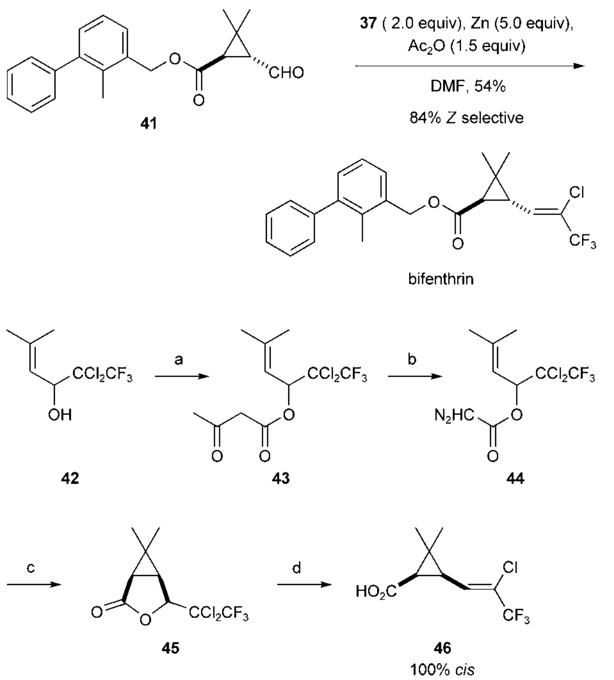
Scheme 10. Generation of CF_3CCl_2ZnCl and addition to aldehydes. DMF = *N*,*N*-dimethylformamide.

The use of zinc in excess (> 2 equiv) in the presence of acetic anhydride in DMF at 50 °C induces reductive β elimination of chloride and acetate ions to afford 2-chloro-1,1,1-trifluoro-2-alkenes **39** in one pot with *Z* selectivity. 2-Chloro-1,1-difluoro-1-alken-3-ols **40** are obtained conveniently by the 1,2-elimination of chloride and fluoride when a catalytic amount of aluminum chloride is employed instead of acetic anhydride (Scheme 11).[29]



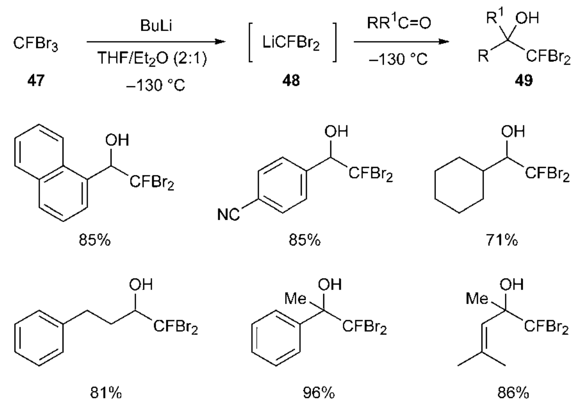
Scheme 11. One-pot synthesis of polyfluorinated alkenes **39** and **40** from CF_3CCl_3.

The synthetic versatility of this method is demonstrated by a one-step synthesis of bifenthrin, in which the non-methyl substituents on the cyclopropane ring are in a *trans* relationship (Scheme 12).[30] In a complementary sequence, the *cis* cyclopropanecarboxylic acid **46** can be formed stereoselectively from the adduct **42** through diazoacetylation, intramolecular cyclopropanation, and reduction with zinc. The present method constitutes a practical approach to the preparation of 3,3,3-trifluoroprop-1-enyl-substituted cyclopropanecarboxylic acids, a key fragment of highly potent artificial pyrethroids.



Scheme 12. Synthesis of bifenthrin and the *cis* cyclopropane **46**: a) diketene, K_2CO_3 (88%); b) TsN_3, NEt_3, then NaOH (82%); c) [Cu(acac)$_2$], dioxane (75%); d) Zn, DMF (84%). acac = acetylacetonate, Ts = *p*-toluenesulfonyl.

In contrast to that of perfluoroalkyl metal reagents, the chemistry of monofluoroalkyl metal reagents still remains to be explored. We succeeded in generating dibromofluoromethyllithium (**48**), which underwent addition to carbonyl derivatives as shown in Scheme 13.[31] The carbenoid reagent

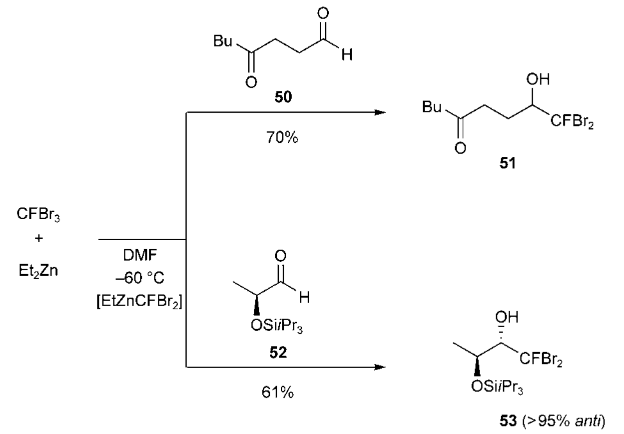


Scheme 13. Generation of $LiCFBr_2$ (**48**) and addition to carbonyl compounds.

48 is prepared by the treatment of tribromofluoromethane (**47**) with butyllithium at −130 °C in a solvent mixture of THF/Et_2O (2:1) in the presence of the aldehyde or ketone substrate. A variety of dibromofluoromethylated alcohols **49** were prepared in high yields in this way. Since **48** reportedly undergoes decomposition through α elimination leading to :CFBr even at −116 °C,[32] its generation at −130 °C in the presence of an electrophile is essential for the success of the transformation.

An alternative reagent to **48** is a zinc carbenoid prepared by bromine–zinc exchange of tribromofluoromethane with diethylzinc in DMF at −60 °C. The zinc carbenoid makes it possible to perform the whole transformation at higher temperatures with high chemoselectivity: Aldehydes are favored significantly over ketones.[33] For example, the ketoaldehyde **50** reacts with the carbenoid to give only the aldehyde adduct **51** (Scheme 14). Diastereoselective addition to the lactaldehyde silyl ether **52** is also possible with the zinc reagent.

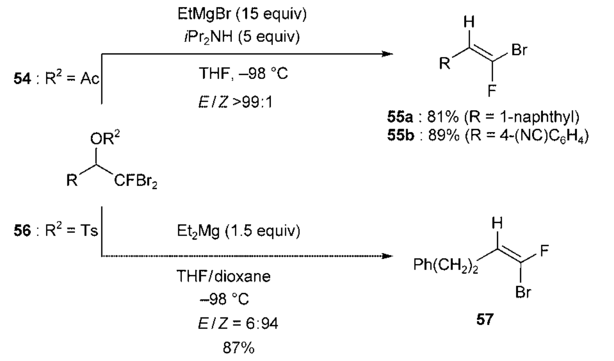
Fluoroolefins play essential roles in liquid-crystalline materials as well as peptide isosteres and enzyme inhibitors.[34] Since the physical properties and biological activity of fluoroolefins greatly depend on configuration, their stereocontrolled synthesis is indispensable. For the stereoselective synthesis of various types of fluoroolefins, stereodefined bromofluoroolefins serve as valuable precursors, as the cross-coupling of bromofluoroolefins with various organometallic reagents in general proceeds to give the target molecules with retention of configuration.[35] Therefore, any new method for



Scheme 14. Chemoselective and stereoselective addition of $EtZnCFBr_2$ to aldehydes.

the facile and stereoselective synthesis of bromofluoroolefins[36] leads to a versatile entry to fluoroolefins.

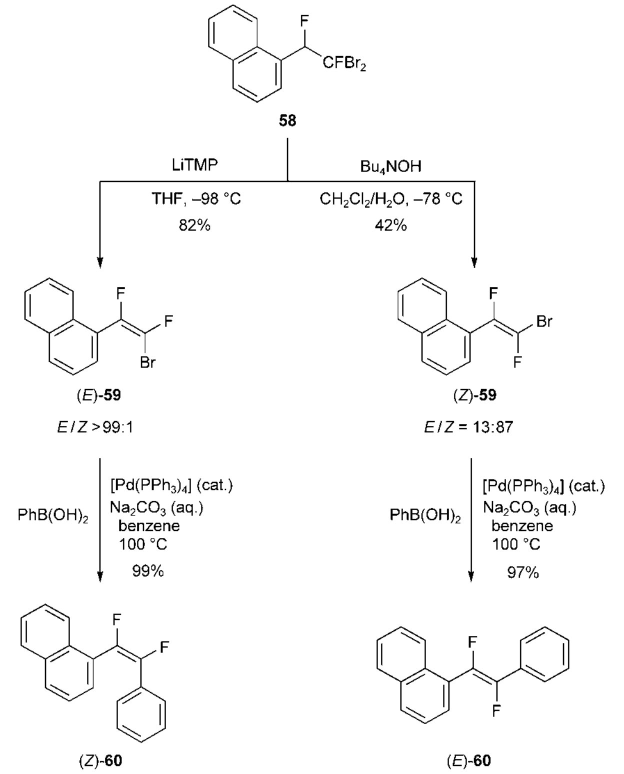
To this end, dibromofluoromethylated alcohols **49** serve as the precursors of bromofluoro- and bromodifluoroolefins. Acetates **54** derived from **49** ($R^1 = H$) are converted into (*E*)-1-bromo-1-fluoro-1-alkenes **55** stereoselectively upon treatment with EtMgBr (15 equiv) and $i\mathrm{Pr_2NH}$ (5 equiv), as shown in Scheme 15. On the other hand, bromine–magnesium exchange of the tosylate **56** ($R = Ph(CH_2)_2$) with Et_2Mg (1.5 equiv) provides **57** with high *Z* selectivity.



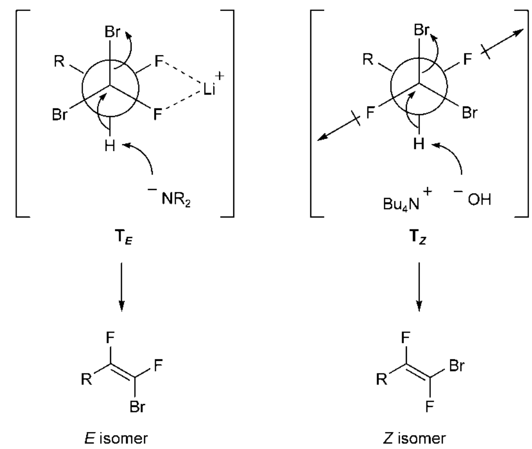
Scheme 15. Stereoselective synthesis of bromofluoroolefins.

The synthesis of 1-bromo-1,2-difluoro-1-alkenes was carried out by dehydrobromination of 1,1-dibromo-1,2-difluoroalkanes, which can be readily prepared from **49** by fluorination with DAST at −78 °C. The treatment of **58** with lithium 2,2,6,6-tetramethylpiperidide (LiTMP) produces (*E*)-**59** as a single diastereomer, whereas (*Z*)-**59** forms preferentially when the metal-free base tetrabutylammonium hydroxide is employed (Scheme 16). These bromides couple with various aryl metal reagents (metal = B, Si, Sn, or Zn): The reaction with $PhB(OH)_2$ is illustrated in Scheme 16.

The stereoselectivity observed in the base-dependent elimination of **58** can be rationalized by a transition-state model that involves chelation of a metal center for *E*-selective elimination and dipole–dipole repulsion of C–F bonds for *Z*-selective elimination with a metal-free base (Scheme 17). In other words, dehydrobromination with LiTMP should proceed via $\mathbf{T}_E$, in which two fluorine atoms are in a synclinal arrangement as a result of Li–F chelation, to produce the



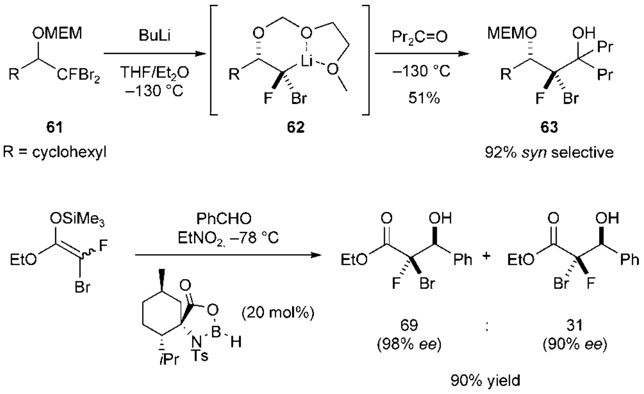
Scheme 16. Stereodivergent synthesis of difluoroolefins.



Scheme 17. Transition-state models for stereoselective dehydrobromination.

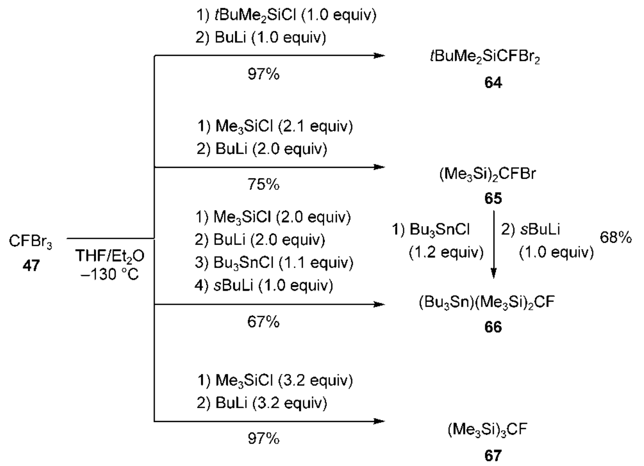
E isomer. On the other hand, the elimination of HBr takes place with Bu_4NOH via $\mathbf{T}_Z$, wherein two C–F bonds prefer to adopt an antiperiplanar arrangement owing to dipole–dipole repulsion, thus giving the *Z* isomer.

Treatment of the MEM (methoxyethoxymethyl) ether **61** with BuLi in THF/Et_2O (2:1) at −130 °C generates the corresponding lithium carbenoid **62**, which can be trapped selectively with 4-heptanone to give the 2-bromo-2-fluoro-1,3-alkanediol derivative **63**.[37] The stereochemical outcome can be explained by the stereoselective generation of the carbenoid **62** through chelation of the lithium ion by the oxygen atoms of the MEM group. The enantioselective synthesis of 2-bromo-2-fluoro-1-hydroxy derivatives is possible by aldol reactions of bromofluoroketene ethyl silyl acetals with aldehydes catalyzed by a chiral Lewis acid (Scheme 18).[38]



Scheme 18. Diastereoselective and enantioselective construction of 2-bromo-2-fluoro-1-alcohols.

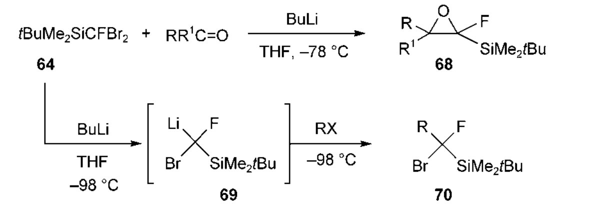
One possibility to improve the thermal stability of fluorine-substituted carbenoids derived from tribromofluoromethane is the introduction of a silyl group at the carbenoid carbon atom, as silyl groups can stabilize an anionic center. Furthermore, the silyl group in the product offers opportunities for functional-group transformations.[39] The fluoromethanes **64**, **65**, and **67** substituted with one, two, or three silyl groups, respectively, are readily prepared by the treatment of $CFBr_3$ (**47**) at −130°C with appropriate stoichiometric amounts of butyllithium in the presence of a chlorosilane (Scheme 19).[40] The fluorobis(silyl)stannylmethane **66** is accessible from **65** through lithiation followed by stannylation. The mixed fluorometallomethane **66** can alternatively



Scheme 19. Preparation of poly(triorganosilyl)methanes from **47**.

be prepared directly from **47** by stepwise silylation and stannylation of the corresponding carbenoids.

The generation of the *tert*-butyldimethylsilyl-substituted carbenoid **69** and its addition to carbonyl compounds is possible at −78°C, thus clearly demonstrating, as expected, that **69** is more stable than **48**, which should be handled at −130°C (Scheme 20). When the reaction mixture is warmed to room temperature, 1-fluoro-1-silyloxiranes **68** are produced in good to excellent yields. No product resulting from

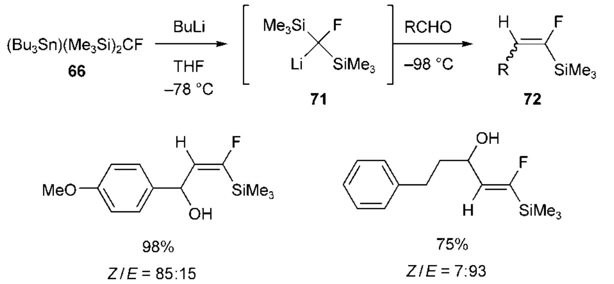


Scheme 20. Addition to carbonyl compounds and alkylation of the silicon-substituted carbenoid **69**.

the Peterson elimination or the Brook rearrangement is formed. The Darzens-type substitution proceeds at the fluorine-substituted carbon atom, a reaction that is thought to be difficult to achieve in an intermolecular sense.[41] Thus, it is evident that the accelerating effect of silicon surpasses the retarding effect of fluorine for the nucleophilic substitution at the α carbon atom.[42] Furthermore, the anion stabilization by silicon allows the addition of an electrophile after the generation of **69** at −98°C, so that alkylation of **69** with alkyl halides is possible.

Whereas bromine–lithium exchange of **65** to generate fluorobis(trimethylsilyl)methyllithium (**71**) in the presence of aldehydes is unsuccessful, tin–lithium exchange of **66** with butyllithium can be performed successfully at −78°C to give **71**, which then reacts with aldehydes to produce 1-fluoro-1-trimethylsilyl-1-alkenes **72** through addition to the carbonyl group followed by a Peterson elimination (Scheme 21).[43] The stereochemical outcome depends on R in the aldehyde RCHO.

The activation of a carbon–silicon bond by a fluoride ion is a highly useful strategy in organic synthesis with organosilicon reagents.[44] This strategy is also highly effective in synthetic fluorine chemistry. For example, trimethyl(trifluoromethyl)silane reacts with a wide variety of electrophiles upon treatment with Bu_4NF (TBAF). Similarly, the treatment of

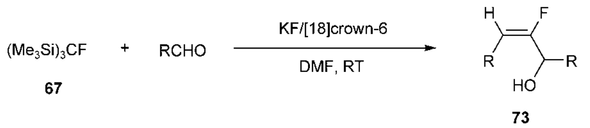


Scheme 21. Synthesis of 1-fluoro-1-silyl-1-alkenes **72**.

(polyfluoroethenyl)silanes with a catalytic amount of tris-(diethylamino)sulfonium difluoro(trimethyl)silicate generates the corresponding naked polyfluoroethenyl anions, which react with aldehydes to produce polyfluorinated allylic alcohols conveniently.[45]

We have found that fluoro(tris(trimethylsilyl))methane (**67**) reacts with 2 molar equivalents of an aromatic aldehyde in the presence of an equimolar amount of KF/[18]crown-6 in DMF at room temperature to give 2-fluoro-1,3-diaryl-2-propen-1-ols **73** in *E*/*Z* ratios of roughly 2:1 (Table 1). The one-pot transformation involves the generation and addition to an aldehyde of the fluorine-substituted naked anion, a Peterson elimination leading to an α-fluoroalkenyl silane, generation of an α-fluoroalkenyl anion, and final aldehyde addition, and thus allows the conversion of three carbon–silicon bonds into a C−C single bond and a C=C double bond in a single operation.

Table 1: Fluoride-mediated addition of **67** to aldehydes.



Entry	R	Yield [%]	*E*/*Z*
1	Ph	74	65:35
2	*p*-tolyl	70	68:32
3	*p*-(MeO)C_6H_4	78	65:35
4	1-naphthyl	65	61:39
5	*p*-PhC_6H_4	68	67:33

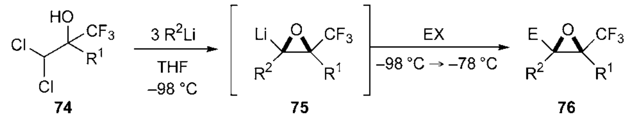
5. Trifluoromethyl-Substituted Oxiranyl Lithium Reagents

Oxiranyl lithium reagents have recently emerged as a highly versatile class of reagents for the stereoselective synthesis of functionalized oxiranes and alkenes.[46] However, fluorine-containing oxiranyl lithium compounds remained unexplored until 2001, when we reported our results on trifluoromethyl-substituted oxiranyl lithium reagents.[47a]

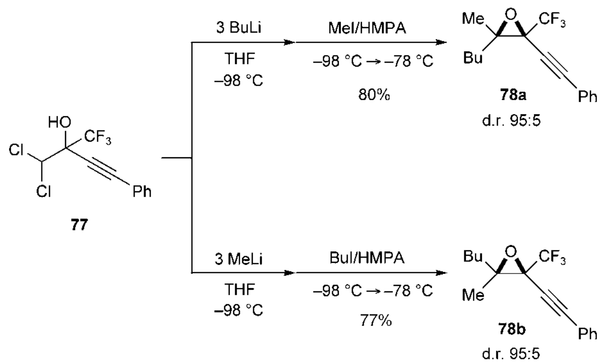
The synthetic elaboration of CF_3-substituted β-oxido carbenoids derived from 3,3-dichloro-1,1,1-trifluoroalkan-2-ols **74** has led to the discovery that oxiranyl lithium intermediates **75** are efficiently produced at −98 °C with 3 equivalents of R^2Li and react with such electrophiles as chlorotrimethylsilane, isopropoxy(pinacolato)borane, and carbonyl compounds, to afford tri- and tetrasubstituted oxiranes **76** in good yields (Table 2).[47] Worthy of emphasis are: 1) the simultaneous nature of oxirane formation and the introduction of the substituent R^2, 2) the high diastereoselectivities observed (d.r. > 95:5) in the formation of **76**, 3) the *cis* arrangement of CF_3 and Li in **75**, and 4) the diastereotopic discrimination between the two chlorine substituents in the acyclic substrate **74**.

With an alkyl halide as the electrophile E−X, the stereodivergent synthesis of tetrasubstituted oxiranes is achieved through the proper choice of R^2Li and E−X. For example, the treatment of **77** with BuLi and then with a solution of iodomethane in HMPA afforded **78a**, whereas the other diastereomer **78b** was obtained by the methyllithium-mediated generation of **75** followed by trapping with 1-iodobutane in HMPA (Scheme 22).

Table 2: Stereoselective synthesis of CF_3-substituted oxiranes.



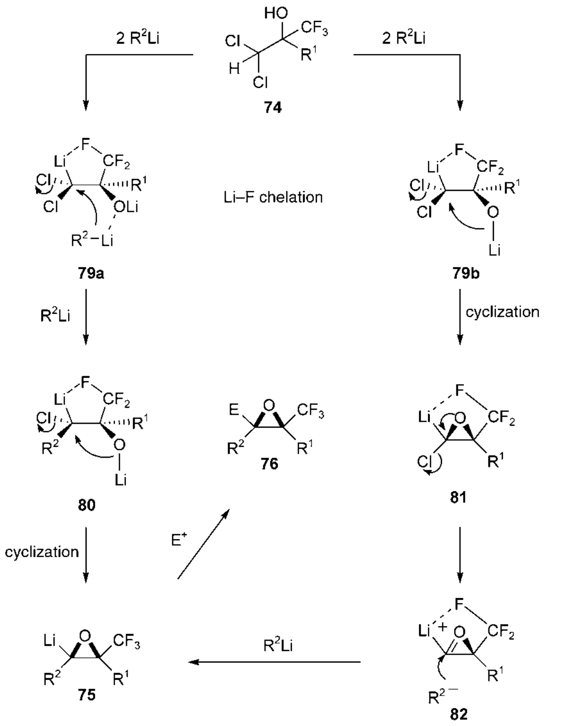
Entry	R^1	R^2	E	Yield [%]	d.r.
1	$Ph(CH_2)_2$	Bu	H	85	> 95:5
2	(*E*)-PhCH=CH	Bu	H	74	83:17
3	PhC≡C	Bu	H	84	> 95:5
4	Ph	Bu	H	83	89:11
5	$Ph(CH_2)_2$	Ph	H	93	> 95:5
6	PhC≡C	Ph	H	95	> 95:5
7	PhC≡C	Bu	$SiMe_3$	79	> 95:5
8	PhC≡C	Ph	$B(OCMe_2)_2$	76	> 95:5
9	$Ph(CH_2)_2$	Bu	$Ph_2C(OH)$	57	> 95:5



Scheme 22. Stereodivergent synthesis of tetrasubstituted oxiranes. HMPA = hexamethyl phosphoramide.

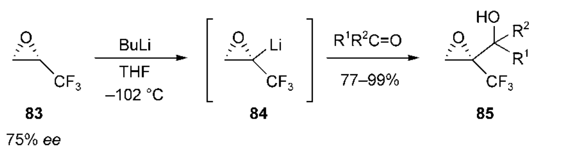
A proposed mechanism for the stereoselective generation of **75** is shown in Scheme 23. At first, it is thought that **74** reacts with 2 equivalents of R^2Li to produce the β-oxido carbenoid **79a** (or **79b**).[48] The assumption of a favored Li–F-chelated conformation as shown in **79a** (or **79b**) leads to two possible pathways: One pathway is the substitution of chlorine in **79a** by R^2Li through coordination with OLi to produce **80**, which should undergo intramolecular cyclization to generate **75**. Intermediate **75** should be stable enough at −98 °C to react with an electrophile and produce **76**. Alternatively, the cyclization of **79b** first gives the chlorolithiooxirane **81**, which might liberate a chloride ion to form the oxyrenium ion **82** by participation of the lone pair of electrons on the oxygen atom. The intermediate **82** is apparently unstable and is trapped immediately by another molecule of R^2Li from the face opposite to Li–F chelation to afford **75**.

Very recently, the trifluoromethyl-substituted oxiranyl anion **84** was generated successfully from optically active 2,3-



Scheme 23. Proposed mechanism for the stereoselective generation of **75**.

epoxy-1,1,1-trifluoropropane (**83**, 75% *ee*) and BuLi in THF at −102 °C (Scheme 24).[49] The configuration of **84** was reported to be stable below −78 °C. Its reaction with aldehydes and ketones gives alcohols **85**.

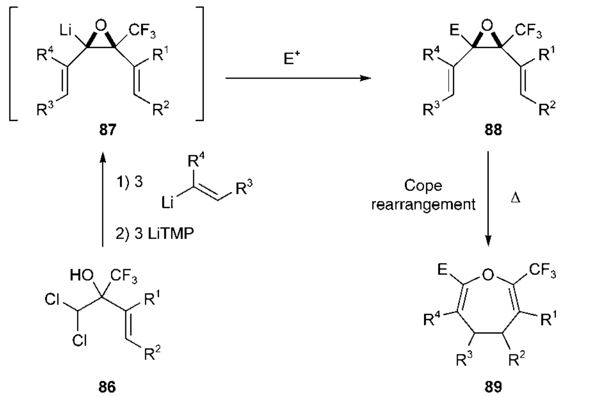


Scheme 24. Generation and reactions of α-CF_3-substituted oxiranyl lithium.

6. A Facile Route to Seven-Membered Oxacycles

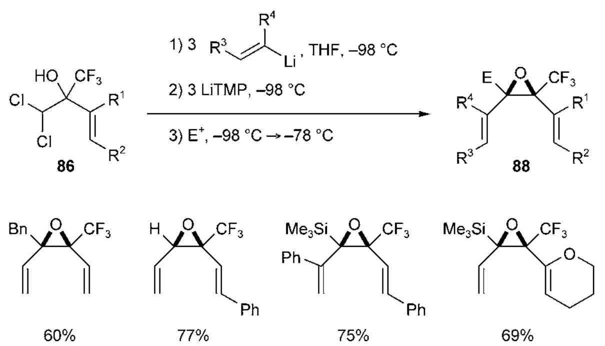
Much attention is focused on seven-membered oxacycles, such as oxepanes and oxepins, owing to their occurrence in biologically active natural products and to valence-bond isomerization between oxepin and benzene oxide.[50] As the incorporation of a CF_3 group into biologically active compounds can alter their properties, facile methods for the synthesis of CF_3-substituted seven-membered oxacycles, a novel class of CF_3-containing compounds, is of particular interest in the quest for potent biologically active substances.

The Cope rearrangement of 2,3-bis(alkenyl)oxiranes would appear to be an attractive approach to the construction of such oxacycles. However, this strategy has remained unrealized, mainly because 1) bis(alkenyl)oxiranes with the essential *cis* arrangement are hard to construct, 2) the rearrangement requires relatively high reaction temperatures, and 3) available disubstituted oxiranes are limited to precursors of 2,7-unsubstituted oxepins. Thus, tri- and tetrasubstituted oxiranes have remained unexplored because of the lack of efficient and stereoselective access routes.[51] We envisaged that it would be possible to prepare 2-CF_3-substituted 2,3-*cis*-bis(alkenyl)oxiranes **88** readily if the protocol for the stereoselective generation of CF_3-substituted oxiranyl lithium compounds was applicable to alkenyl-substituted dichlorohydrins **86** and alkenyl lithium intermediates (Scheme 25).



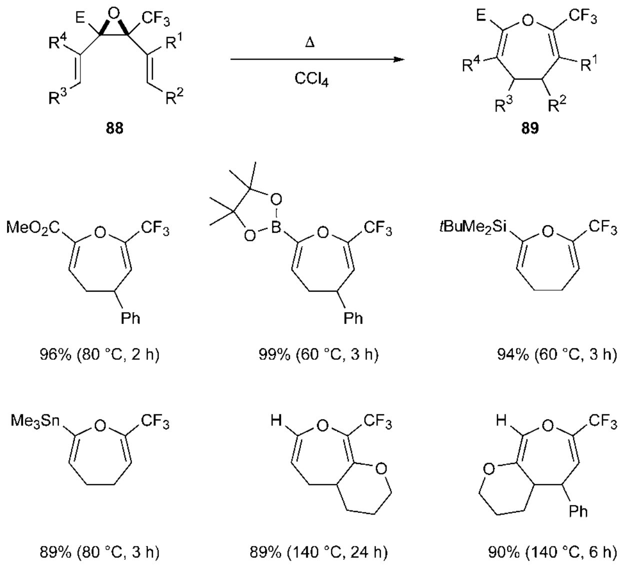
Scheme 25. Synthetic plan for the preparation of 2-trifluoromethyl-4,5-dihydrooxepins.

Regretfully, however, no trace of the desired reaction took place under the original conditions, probably as a result of the low basicity of the alkenyl lithium intermediates and/or the low acidity of a methine proton of **86**. After many trials, we found that the addition of LiTMP (3 equiv) to a solution of **86** and an alkenyl lithium compound is a highly effective way to generate oxiranyl lithium intermediates **87**, which can be trapped with a variety of electrophiles (E^+) to give **88** with excellent *cis* selectivity (Scheme 26).[52]



Scheme 26. Synthesis of *cis* bis(alkenyl)oxiranes **88**.

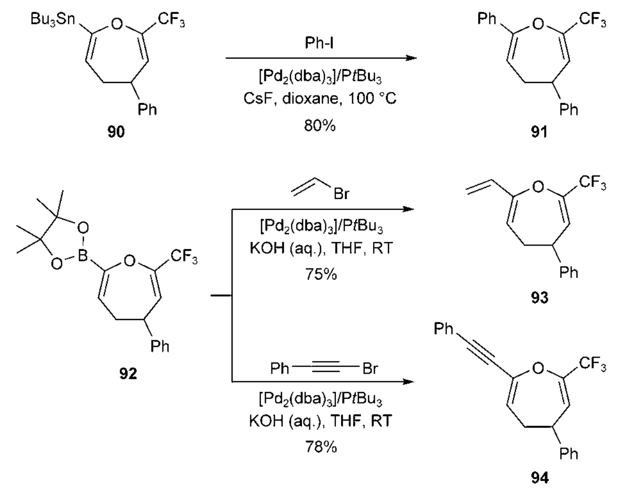
The Cope rearrangement of **88** proceeded smoothly upon heating at 60–140 °C to afford **89** in good to excellent yields (Scheme 27).[52] The rate of the reaction depends on the substituent E. If E is bulky, it appears to force the two



Scheme 27. Cope rearrangement of **88**.

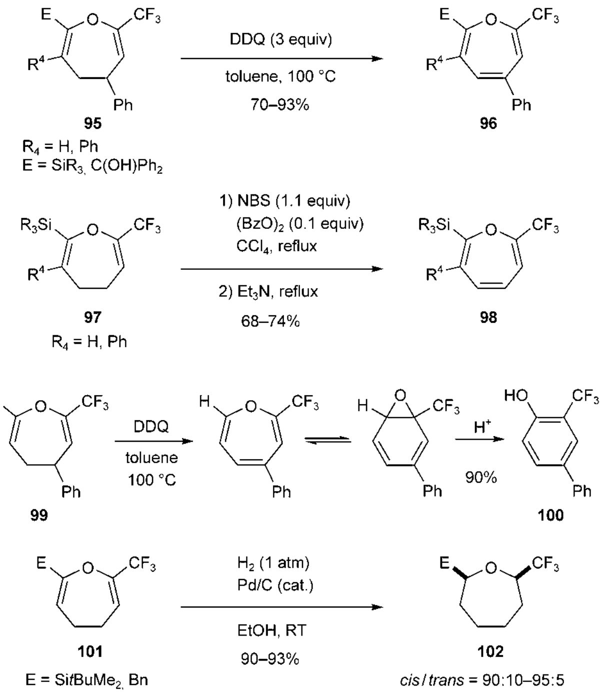
terminal C(sp^2) atoms close together, thus accelerating the rearrangement. If the starting substrate contains a cyclic olefin, oxepins with a fused framework, which are otherwise difficult to prepare, are obtained.

An additional feature of the present approach is the versatility of products containing metalloid elements introduced during the preparation of oxiranes **88**. Thus, a tin or boron substituent is readily converted into an aryl, alkenyl, or alkynyl group by a Pd-catalyzed cross-coupling reaction with an aryl, alkenyl, or alkynyl halide, respectively (Scheme 28).



Scheme 28. Coupling reaction of metal-substituted oxepins.

Dihydrooxepins **95** and **97** are readily converted into oxepins **96** (by oxidation with DDQ) and **98** (by allylic bromination–dehydrobromination with NBS and NEt_3), respectively, as illustrated in Scheme 29.[52] Oxidation of the 7-unsubstituted 4,5-dihydrooxepin **99** with DDQ produces **100** through equilibration of the oxepin to a benzene oxide, followed by acid-induced isomerization. Finally, 2-CF_3-substituted oxepanes **102** are readily available through the catalytic hydrogenation (1 atm) of **101** in the presence of 5% Pd/C.



Scheme 29. Oxidation and reduction of 2-CF_3-4,5-dihydrooxepins. Bn = benzyl, Bz = benzoyl, DDQ = 2,3-dichloro-5,6-dicyano-1,4-benzoquinone.

7. Stereoselective Synthesis of CF_3-Containing Tetrasubstituted Alkenes

The regio- and stereoselective synthesis of tetrasubstituted alkenes is always a challenging issue in organic synthesis.[53] In particular, fluorine-containing tetrasubstituted alkenes are crucial in the search for potential fluorinated drugs. For example, panomifene is reported to exhibit antiestrogenic activity superior to that of tamoxifen, a widely used drug for the treatment of breast cancer (Figure 5).[54]

As oxiranyl lithium compounds are known to react with organometallic reagents, such as RLi, R_3Al, R_2Zn, and R_4CeLi, to give alkylated alkenes,[55] we first attempted unsuccessfully to apply this methodology to **103**. Fortunately,

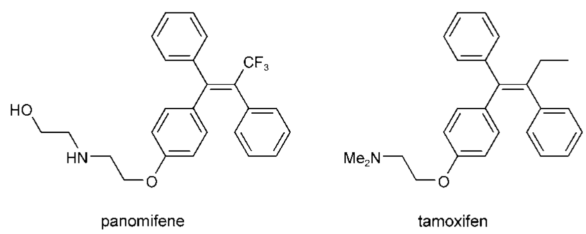
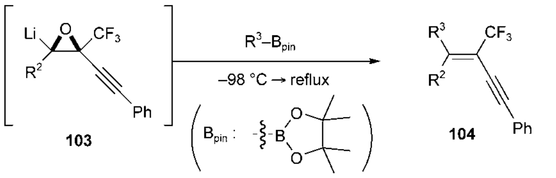


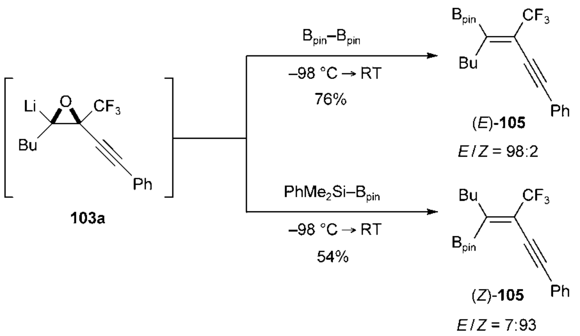
Figure 5. Examples of nonsteroidal antiestrogens.

we have since found that the treatment of **103** with organo(pinacolato)borane reagents R^3–B_{pin} (pin = pinacolato) affords tetrasubstituted alkenes stereoselectively in good yields, with CF_3 and R^3 in a *cis* relationship (Table 3). An appropriate combination of R^2 and R^3 allows the preparation of either stereoisomer at will (entries 3 and 4 of Table 3). The stereodivergent synthesis of β-CF_3-substituted alkenyl boronates **105** is also possible when bis(pinacolato)diboron or dimethylphenylsilyl(pinacolato)boron is employed with **103**. The stereochemical outcome with the diboron reagent is opposite to that observed with the silylborane and the same as that with R^3–B_{pin} (Scheme 30).

Table 3: Stereoselective synthesis of CF_3-containing tetrasubstituted alkenes.

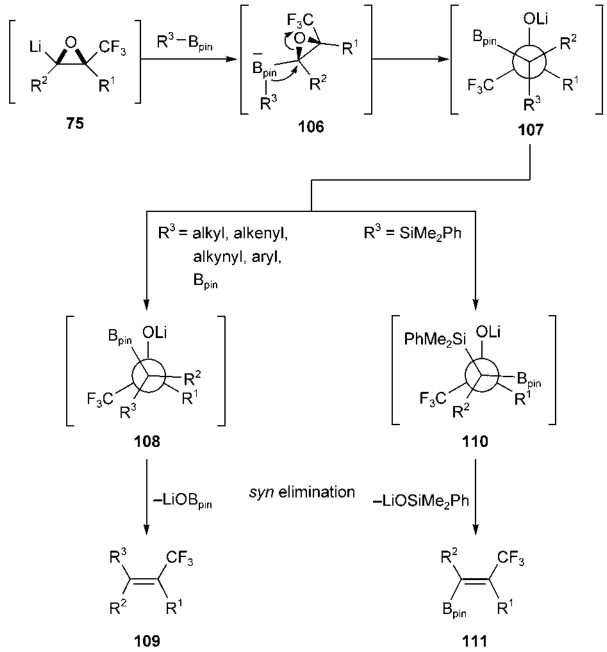


Entry	R^2	R^3	Yield [%]	d.r.
1	Bu	PhC≡C	74	> 95:5
2	Bu	(*E*)-BuCH=CH	56	> 95:5
3	Bu	Ph	66	97:3
4	Ph	Bu	84	96:4



Scheme 30. Stereoselective synthesis of CF_3-containing tetrasubstituted alkenes.

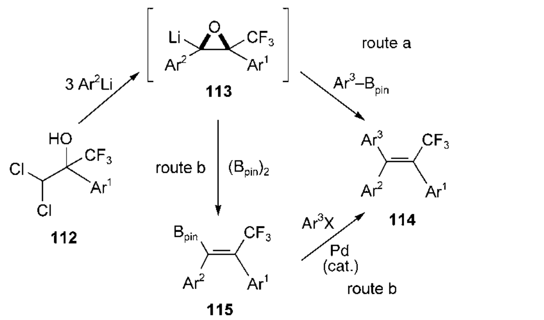
The stereoselective and stereodivergent formation of (*E*)- or (*Z*)-**105** can be explained by assuming that a *syn* elimination of $LiOB_{pin}$ or $LiOSiPhMe_2$ takes place in the last step, as illustrated in Scheme 31. First, **75** reacts with the borane reagent R^3–B_{pin} to give the borate **106**, which undergoes 1,2-migration of the substituent R^3 and opening of the oxirane to generate the lithium alkoxide **107**.[56] If R^3 is an alkyl, alkenyl, alkynyl, aryl, or boryl group, minimal rotation is required to attain the conformation **108**, in which the carbon–boron and carbon–oxygen bonds are eclipsed. The alkene **109** is then produced by the *syn* elimination of $LiOB_{pin}$.[57] When the silylborane is employed (i.e. R^3 in **106** is the silyl group), a rotation of 180° would be required for $LiOSiMe_2Ph$ to be eliminated in a *syn* fashion to give **111**. The elimination of



Scheme 31. Mechanism for stereoselective alkene formation.

$LiOSiMe_2Ph$ rather than $LiOB_{pin}$ might be expected, as a $PhMe_2Si$ group is more oxophilic than a B_{pin} group.

Based on these results, it seemed that various CF_3-substituted triaryl ethenes **114**, including panomifene, would be accessible if diaryl-substituted oxiranyl lithium intermediates **113** are generated successfully from Ar^1-substituted dichlorohydrins in the presence of Ar^2Li and react with Ar^3–B_{pin} smoothly (route a in Scheme 32).[58] However, this

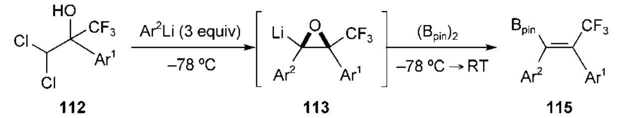


Scheme 32. Synthetic plan for the preparation of CF_3-substituted triaryl ethenes.

expectation was not fulfilled, although it was shown that intermediate **113** was generated quite efficiently even at −78 °C. Hence, we took another approach (route b), which involves the preparation and cross-coupling of β-CF_3-substituted alkenyl boron reagents **115**.[59] Thus, the lithiooxirane **113** is first generated from **112** with Ar^2Li at −78 °C and then allowed to react with the diboron reagent to produce **115** in good yields with high *E* selectivities (Table 4).

The Pd-catalyzed cross-coupling reaction of **115** (route b in Scheme 32) with a variety of aryl iodides was carried out

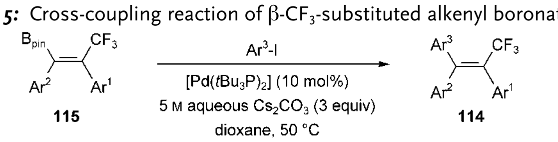
Table 4: Stereoselective preparation of β-CF_3-substituted alkenyl boronates.



Ar^1	Ar^2	Yield [%]	*E*/*Z*
Ph	Ph	63	98:2
Ph	*p*-ClC_6H_4	73	97:3
Ph	*p*-$(MeO)C_6H_4$	78	94:6
Ph	*p*-$(MOMO)C_6H_4$	75	92:8
p-ClC_6H_4	Ph	80	99:1
p-$(MeO)C_6H_4$	Ph	63	99:1

successfully in the presence of a 5 M aqueous solution of Cs_2CO_3 to give **114** as a single stereoisomer in high to excellent yields (Table 5).[59] The coupling reaction under

Table 5: Cross-coupling reaction of β-CF_3-substituted alkenyl boronates.



	Ar^1	Ar^2	Ar^3	Yield [%]
114a	Ph	Ph	*p*-$(EtO_2C)C_6H_4$	92
114b	Ph	Ph	*p*-FC_6H_4	91
114c	Ph	Ph	*p*-ClC_6H_4	96
114d	Ph	*p*-$(MeO)C_6H_4$	Ph	90
114e	Ph	*p*-ClC_6H_4	*p*-$(MeO)C_6H_4$	89
114f	Ph	*p*-$(MOMO)C_6H_4$	Ph	92
114g	*p*-$(MeO)C_6H_4$	Ph	*p*-FC_6H_4	94

anhydrous conditions causes deborylation of **115** to some extent. Neither electron-donating nor electron-withdrawing groups on Ar^1 or Ar^2 affect the reaction. The demethylation of **114d** with NaOEt followed by the introduction of a 2-(2-hydroxyethyl)aminoethyl moiety completed a facile synthesis of panomifene.[59]

8. Asymmetric Synthesis of 1,1,1-Trifluoro-2-alkanols

In recent years, many optically active organofluorine compounds have been employed in the pharmaceutical, agrochemical, and materials sciences. For example, many antiferroelectric liquid crystals contain optically active 1,1,1-trifluoro-2-alkanols as key fragments that contribute to the stabilization of the SmC_A phase (Figure 6). Accordingly, the enantioselective synthesis of such fluorinated alcohols has been a target in organic synthesis.[60] We have designed novel methods for the efficient synthesis of 1,1,1-trifluoro-2-alkanols, including the asymmetric synthesis of trifluoromethyl-substituted hemiacetals and their stereospecific nucleophilic substitution.[61]

In our studies on asymmetric hemiacetal synthesis we found that bubbling gaseous trifluoroacetaldehyde, generated from 1-ethoxy-2,2,2-trifluoroethanol (**118**) and polyphos-

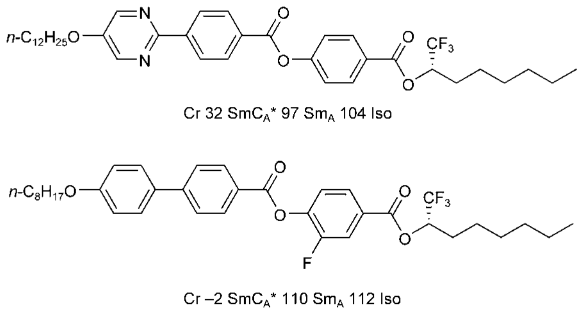
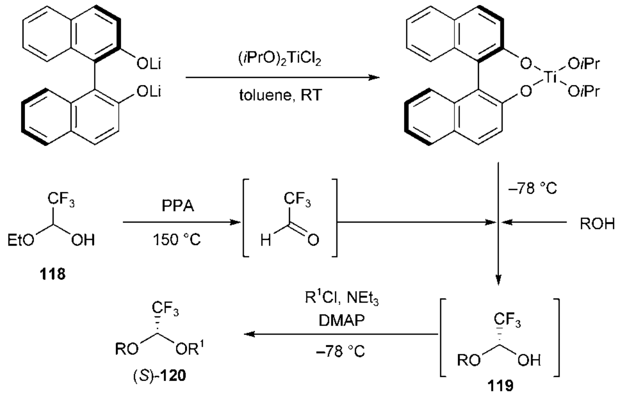


Figure 6. Examples of antiferroelectric liquid crystals.

phoric acid (PPA), into a solution of an alcohol in toluene in the presence of a preprepared (*R*)-binol–Ti(O*i*Pr)$_2$ catalyst yielded the optically active hemiacetal **119**, which then underwent acylation with an acid chloride or sulfonyl chloride at low temperatures to give the optically active mixed acetal (*S*)-**120** (Table 6).[62] Alternatively, optically pure 1-alkoxy-

Table 6: Asymmetric synthesis of CF_3-substituted hemiacetals.



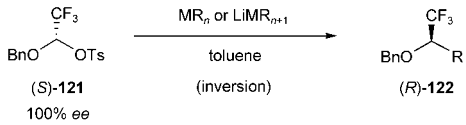
Entry	R	R^1	Yield [%]	*ee* [%]
1	Bu	PhCO	65	65
2	$Ph(CH_2)_2$	PhCO	57	78
3	*i*Pr	PhCO	85	85
4	Et_2CH	PhCO	54	73
5	$PhCH_2$	PhCO	54	91
6	$PhCH_2$	$MeSO_2$	35	79

DMAP = 4-dimethylaminopyridine.

2,2,2-ethyl tosylates are accessible by the optical resolution of racemic tosylates by chiral preparative HPLC (Daicel chiralpak AD or chiralcel OD).[63] The presence of the CF_3 group makes it possible to isolate otherwise unstable α-alkoxyalkyl tosylates.

Hemiacetal tosylates (*S*)-**121** react with organoaluminum, organoboron, and organozinc reagents to give substitution products with inversion of configuration.[64] The best overall results with respect to the yields and *ee* values of the isolated products were obtained with aluminate reagents (Table 7). This study demonstrated for the first time that intermolecular nucleophilic substitution is possible at a CF_3-substituted carbon atom to form a carbon–carbon bond. We should emphasize at this point that the alkoxy substituent in **121** assists in this process by increasing the reactivity of the

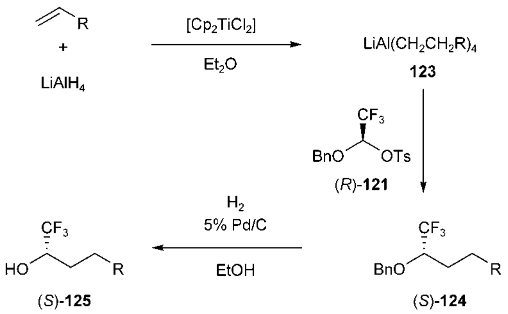
Table 7: Nucleophilic substitution of hemiacetal **121** with organometallic reagents.



Entry	MR_n or $LiMR_{n+1}$	T [°C]	Yield [%]	*ee* [%]
1	$AlEt_3$	−20	75	82
2	$LiAlEt_4$	0	69	98
3	$LiAlBu_4$	RT	62	90
4	$LiAl(iBu)_4$	0	46	93
5	$LiBEt_4$	RT	32	95
6	$LiZnEt_3$	RT	30	96

otherwise unreactive CF_3-substituted alkyl tosylates. Thus, the presence of the substituent OCH_2Ph α to a CF_3 group in **121** is essential for the success of the nucleophilic substitution. The intramolecular nucleophilic substitution of 1,1,1-trifluoro-2-alkanol derivatives with an amino group or a cyano-stabilized carbanion is reported to take place with inversion of configuration, giving rise to trifluoromethyl-substituted aziridines and cyclopropanes, respectively.[65]

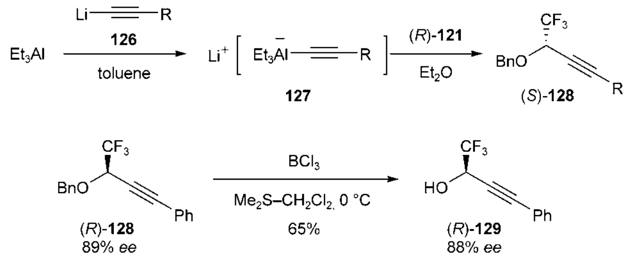
The versatility of the present approach is demonstrated by the preparation of 1,1,1-trifluoro-2-alkanols (Scheme 33).



Scheme 33. Efficient route to optically active 1,1,1-trifluoro-2-alkanols.

Thus, the titanocene-catalyzed hydroalumination of 1-alkenes with $LiAlH_4$ produces regioselectively lithium tetraalkyl aluminates **123**, which undergo stereospecific substitution with (*R*)-**121** to give (*S*)-**124**. The hydrogenolysis of (*S*)-**124** under a hydrogen atmosphere in the presence of 5% Pd/C gives optically active alcohols (*S*)-**125**. Any alkanol of the type **125** can be prepared from the corresponding readily available 1-alkene and the tosylate **121** by this sequence.[66]

As only one alkyl group in aluminates takes part in the substitution, the use of mixed aluminates that transfer a required moiety to the exclusion of the less reactive groups is more practical. Indeed, 1-alkynyl trialkyl aluminum compounds are appropriate for this purpose.[67] Thus, mixed aluminates **127**, prepared from commercially available triethylaluminum and the requisite alkynyl lithium reagent **126**, undergo reaction with (*R*)-**121** selectively and stereospecifically to afford alkynylated products (*S*)-**128** with inversion of configuration (Scheme 34). The catalytic hydrogenation of (*R*)-**128** at 1 atm with PtO_2 furnishes the benzyl ether (*R*)-**124**



Scheme 34. Alkynylation of the optically active tosylate (*R*)-**121** with the mixed aluminate **127**.

(R = Ph), whereas debenzylation of (*R*)-**128** with BCl_3 and Me_2S gives the alcohol (*R*)-**129**, leaving both the CF_3 and C≡C functional groups untouched.

9. Summary

Herein we have reviewed our progress during the last two decades in synthetic fluorine chemistry. A fluorination method (oxidative desulfurization–fluorination) has been developed, as well as building-block methods in which readily available CF_3CCl_3, $CFBr_3$, $CF_3COCHCl_2$, $CF_3CH(OH)OEt$, and their derivatives are used in combination with organometallic reagents. These novel synthetic methods have allowed the convenient preparation of a variety of fluorine-containing target molecules. The presence of fluorinated or perfluorinated substituents often induces unusual phenomena, which may be ascribed ultimately to the strong electron-withdrawing effect of these substituents. From this point of view, insight into synthetic fluorine chemistry opens possibilities for the design of new reactions/reagents not only in organofluorine chemistry but also in general organic synthesis and organometallic chemistry. We hope this Review may foster research in related fields.

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[1] a) T. Hiyama, *Organofluorine Compounds: Chemistry and Applications*, Springer, Berlin, **2000**; b) *Houben-Weyl, Vol. E10a* (Eds.: B. Baasner, H. Hagemann, J. C. Tatlow), Thieme, Stuttgart, **2000**; c) *Biomedical Frontiers of Fluorine Chemistry*, *ACS Symposium Series 639* (Eds.: I. Ojima, J. R. McCarthy, J. T. Welch), American Chemical Society, Washington, **1996**; d) *Chemistry of Organic Fluorine Compounds II: A Critical Review*, *ACS Monograph 187* (Eds.: M. Hudlicky, A. E.

Pavlath), American Chemical Society, Washington, **1995**; e) *Organofluorine Chemistry—Principles and Commercial Applications* (Eds.: R. E. Banks, B. E. Smart, J. C. Tatlow), Plenum Press, New York, **1994**; f) *Chemistry of Organic Fluorine Compounds*, 2nd ed. (Ed.: M. Hudlicky), Ellis Horwood, New York, **1992**; g) *Selective Fluorination in Organic and Bioorganic Chemistry, ACS Symposium Series 456* (Ed.: J. T. Welch), American Chemical Society, Washington, **1991**; h) *Fluorine-containing Molecules: Structure, Reactivity, Synthesis, and Applications* (Eds.: J. F. Liebman, A. Greenberg, J. W. R. Dolbier), VCH, New York, **1988**; i) J. T. Welch, *Tetrahedron* **1987**, *43*, 3123; j) M. Schlosser, *Tetrahedron* **1978**, *34*, 3; k) P. Kirsch, *Modern Fluoroorganic Chemistry*, Wiley-VCH, Weinheim, **2004**.

[2] B. E. Smart, *J. Fluorine Chem.* **2001**, *109*, 3.

[3] Special issue on "Fluorine in the Life Sciences", *ChemBioChem* **2004**, *5*, 557–726.

[4] M. Schlosser, *Angew. Chem.* **1998**, *110*, 1538; *Angew. Chem. Int. Ed.* **1998**, *37*, 1496.

[5] D. A. Dixon, T. Fukunaga, B. E. Smart, *J. Am. Chem. Soc.* **1986**, *108*, 4027.

[6] For C−H···F interactions, see: a) J. A. K. Howard, V. J. Hoy, D. O'Hagan, G. T. Smith, *Tetrahedron* **1996**, *52*, 12613; b) J. D. Dunitz, R. Taylor, *Chem. Eur. J.* **1997**, *3*, 89; c) I. Hyla-Kryspin, G. Haufe, S. Grimme, *Chem. Eur. J.* **2004**, *10*, 3411; d) J. D. Dunitz, *ChemBioChem* **2004**, *5*, 614; for C−M···F interactions, see: e) H. Plenio, *Chem. Rev.* **1997**, *97*, 3363; f) H. Plenio, *ChemBioChem* **2004**, *5*, 65.

[7] a) Reference [1b]; b) *Houben-Weyl*, *Vol. E10b/1* (Eds.: B. Baasner, H. Hagemann, J. C. Tatlow), Thieme, Stuttgart, **2000**; c) *Houben-Weyl*, *Vol. E10b/2* (Eds.: B. Baasner, H. Hagemann, J. C. Tatlow), Thieme, Stuttgart, **2000**; d) T. Kitazume, T. Yamazaki, *Experimental Methods in Organic Fluorine Chemistry*, Kodansha, Tokyo, **1998**; e) *Top. Curr. Chemi.* **1997**, *192*; f) *Top. Curr. Chem.* **1997**, *193*; g) *Synthetic Fluorine Chemistry* (Eds.: G. A. Olah, R. D. Chambers, G. K. S. Prakash), Wiley, New York, **1992**.

[8] a) Reference [7a,b]; b) J. Hutchinson, G. Sandford, *Top. Curr. Chem.* **1997**, *193*, 1; c) J. A. Wilkinson, *Chem. Rev.* **1992**, *92*, 505.

[9] a) Reference [7b,c]; b) J. M. Percy, *Top. Curr. Chem.* **1997**, *193*, 131.

[10] For a review on halofluorination, see: a) *Houben-Weyl*, *Vol. E10a* (Eds.: B. Baasner, H. Hagemann, J. C. Tatlow), Thieme, Stuttgart, **2000**, p. 122; b) G. A. Olah, J. T. Welch, Y. D. Vankar, M. Nojima, I. Kerekes, J. A. Olah, *J. Org. Chem.* **1979**, *44*, 3872; c) C. York, G. K. S. Prakash, G. A. Olah, *J. Org. Chem.* **1994**, *59*, 6493; d) M. Kuroboshi, T. Hiyama, *Bull. Chem. Soc. Jpn.* **1995**, *68*, 1799.

[11] a) G. Haufe, *Tetrahedron Lett.* **1988**, *29*, 2311; b) C. Saluzzo, G. Alvernhe, D. Anker, G. Haufe, *Tetrahedron Lett.* **1990**, *31*, 663.

[12] See reference [10b].

[13] M. Kuroboshi, K. Kanie, T. Hiyama, *Adv. Synth. Catal.* **2001**, *343*, 235.

[14] S. Furuta, M. Kuroboshi, T. Hiyama, *Bull. Chem. Soc. Jpn.* **1999**, *72*, 805.

[15] For a review on the synthesis of *gem*-difluoromethylene compounds, see: M. J. Tozer, T. F. Herpin, *Tetrahedron* **1996**, *52*, 8619.

[16] M. Kuroboshi, T. Hiyama, *Synlett* **1991**, 909.

[17] a) W. B. Motherwell, J. A. Wilkinson, *Synlett* **1991**, 191; b) T. Fuchigami, T. Fujita, *J. Org. Chem.* **1994**, *59*, 7190.

[18] M. Kuroboshi, T. Hiyama, unpublished results. The Friedel–Crafts acylation of 1,2-dimethoxybenzne with **17**, leading to **18**, is reported independently: K. Burger, M. Rudolph, H. Neuhauser, *Liebigs Ann. Chem.* **1991**, 1365.

[19] a) M. Kuroboshi, T. Hiyama, *Synlett* **1994**, 251; see also: b) G. S. Lal, E. Lobach, A. Evans, *J. Org. Chem.* **2000**, *65*, 4830.

[20] K. Kanie, Y. Tanaka, K. Suzuki, M. Kuroboshi, T. Hiyama, *Bull. Chem. Soc. Jpn.* **2000**, *73*, 471.

[21] M. J. Koen, F. I. Guyader, W. B. Motherwell, *J. Chem. Soc. Chem. Commun.* **1995**, 1241.

[22] K. Kanie, S. Takehara, T. Hiyama, *Bull. Chem. Soc. Jpn.* **2000**, *73*, 1875.

[23] a) K. Kanie, K. Mizuno, M. Kuroboshi, T. Hiyama, *Bull. Chem. Soc. Jpn.* **1998**, *71*, 1973; b) P. Kirsch, M. Bremer, *Angew. Chem.* **2000**, *112*, 4384; *Angew. Chem. Int. Ed.* **2000**, *39*, 4216.

[24] For reviews on fluorinated organometallic compounds, see: a) D. J. Burton, Z.-Y. Yang, *Tetrahedron* **1992**, *48*, 189; b) D. J. Burton, Z.-Y. Yang, P. A. Morken, *Tetrahedron* **1994**, *50*, 2993; c) D. J. Burton, L. Lu, *Top. Curr. Chem.* **1997**, *193*, 45.

[25] For a Review on the synthesis of pyrethroid acids, see: D. Arlt, M. Jautelat, R. Lantzsch, *Angew. Chem.* **1981**, *93*, 719; *Angew. Chem. Int. Ed. Engl.* **1981**, *20*, 703.

[26] a) M. Fujita, T. Morita, T. Hiyama, *Tetrahedron Lett.* **1986**, *27*, 2135; b) M. Fujita, T. Hiyama, *Bull. Chem. Soc. Jpn.* **1987**, *60*, 4377.

[27] a) R. W. Lang, *Helv. Chim. Acta* **1986**, *69*, 881; see also: b) R. W. Lang, *Helv. Chim. Acta* **1988**, *71*, 369.

[28] D. Bellus, B. Klingert, R. W. Lang, G. Rihs, *J. Organomet. Chem.* **1988**, *339*, 17.

[29] a) M. Fujita, T. Hiyama, *Tetrahedron Lett.* **1986**, *27*, 3655; b) M. Fujita, K. Kondo, T. Hiyama, *Bull. Chem. Soc. Jpn.* **1987**, *60*, 4385.

[30] a) M. Fujita, T. Hiyama, K. Kondo, *Tetrahedron Lett.* **1986**, *27*, 2139; b) reference [29b].

[31] a) M. Kuroboshi, N. Yamada, Y. Takebe, T. Hiyama, *Synlett* **1995**, 987; b) M. Shimizu, N. Yamada, Y. Takebe, T. Hata, M. Kuroboshi, T. Hiyama, *Bull. Chem. Soc. Jpn.* **1998**, *71*, 2903.

[32] D. J. Burton, J. L. Hahnfeld, *J. Org. Chem.* **1977**, *42*, 828.

[33] a) T. Hata, M. Shimizu, T. Hiyama, *Synlett* **1996**, 831; b) T. Hata, H. Kitagawa, M. Shimizu, T. Hiyama, *Bull. Chem. Soc. Jpn.* **2000**, *73*, 1691.

[34] a) O. Yokokoji, T. Shimizu, S. Kumai, JP 08040952, **1996** [*Chem. Abstr.* **1996**, *124*, 316586]; b) T. Allmendinger, E. Felder, E. Hungerbühler in *Selective Fluorination in Organic and Bioorganic Chemistry* (Ed.: J. T. Welch), American Chemical Society, Washington, **1991**, p. 186; c) P. Bey, J. R. McCarthy, I. A. McDonald in *Selective Fluorination in Organic and Bioorganic Chemistry* (Ed.: J. T. Welch), American Chemical Society, Washington, **1991**, p. 105.

[35] a) *Metal-catalyzed Cross-coupling Reactions* (Eds.: F. Diederich, P. J. Stang), Wiley-VCH, Weinheim, **1998**; b) *Top. Curr. Chem.* **2002**, *219*.

[36] a) V. A. Petrov, C. G. Krespan, B. E. Smart, *J. Fluorine Chem.* **1996**, *77*, 139; b) J. Weber, L. Xu, U. H. Brinker, *Tetrahedron Lett.* **1992**, *33*, 4537; c) S. Eddarir, C. Francesch, H. Mestdagh, C. Rolando, *Tetrahedron Lett.* **1990**, *31*, 4449; d) T. Gouyon, R. Sauvetre, J. F. Normant, *J. Organomet. Chem.* **1990**, *394*, 37; e) M. Shimizu, G.-H. Cheng, H. Yoshioka, *J. Fluorine Chem.* **1988**, *41*, 425; f) T. Fuchikami, I. Ojima, *J. Organomet. Chem.* **1981**, *212*, 145.

[37] a) M. Shimizu, Y. Takebe, M. Kuroboshi, T. Hiyama, *Tetrahedron Lett.* **1996**, *37*, 7387; b) reference [31b].

[38] K. Iseki, Y. Kuroki, Y. Kobayashi, *Tetrahedron Lett.* **1997**, *38*, 7209.

[39] a) A. R. Bassindale, P. G. Taylor in *The Chemistry of Organic Silicon Compounds*, *Vol. 2* (Eds.: S. Patai, Z. Rappoport), Wiley, New York, **1989**, p. 893; b) J. S. Panek in *Comprehensive Organic Synthesis*, *Vol. 1* (Eds.: B. M. Trost, I. Fleming), Pergamon, London, **1991**, p. 579.

[40] a) M. Shimizu, T. Hata, T. Hiyama, *Tetrahedron Lett.* **1997**, *38*, 4591; b) M. Shimizu, T. Hata, T. Hiyama, *Tetrahedron Lett.* **1999**, *40*, 7375; c) M. Shimizu, T. Hata, T. Hiyama, *Heterocycles* **2000**, *52*, 707; d) M. Shimizu, T. Hata, T. Hiyama, *Bull. Chem. Soc. Jpn.*

2000, *73*, 1685; see also: e) H. Burger, P. Moritz, *Organometallics* **1993**, *12*, 4930.

[41] B. E. Smart in *Organofluorine Chemistry—Principles and Commercial Applications* (Eds.: R. E. Banks, B. E. Smart, J. C. Tatlow), Plenum, New York, **1994**, p. 57.

[42] a) I. Fleming in *Comprehensive Organic Chemistry*, *Vol. 3* (Eds.: D. Barton, W. D. Ollis), Pergamon, Oxford, **1979**, p. 541; b) A. R. Bassindale, P. G. Taylor, in *The Chemistry of Organic Silicon Compounds*, *Vol. 2* (Eds.: S. Patai, Z. Rappoport), Wiley, New York, **1989**, p. 951.

[43] For examples of vinylsilane products from bis- or tris(triorganosilyl)methanes, see: a) H. Sakurai, K.-i. Nishiwaki, M. Kira, *Tetrahedron Lett.* **1973**, 4193; b) B.-T. Gröbel, D. Seebach, *Angew. Chem.* **1974**, *86*, 102; *Angew. Chem. Int. Ed. Engl.* **1974**, *13*, 83; c) B.-T. Grobel, D. Seebach, *Chem. Ber.* **1977**, *110*, 852; d) D. Seyferth, J. L. Lefferts, J. R. L. Lambert, *J. Organomet. Chem.* **1977**, *142*, 39; e) I. Fleming, C. D. Floyd, *J. Chem. Soc. Perkin Trans. 1* **1981**, 969; f) D. J. Ager, *J. Org. Chem.* **1984**, *49*, 168; g) P. F. Hudrlik, E. L. O. Agwaramgbo, A. M. Hudrlik, *J. Org. Chem.* **1989**, *54*, 5613; h) C. Palomo, J. M. Aizpurua, J. M. Garcia, I. Ganboa, F. P. Cossio, B. Lecea, C. Lopez, *J. Org. Chem.* **1990**, *55*, 2498.

[44] a) G. G. Furin, O. A. Vyazankina, B. A. Gostevsky, N. S. Vyazankin, *Tetrahedron* **1988**, *44*, 2675; b) T. Hiyama, E. Shirakawa, *Top. Curr. Chem.* **2002**, *219*, 61; c) T. Hiyama in *Metal-catalyzed Cross-coupling Reactions* (Eds.: F. Diederich, P. J. Stang), Wiley-VCH, Weinheim, **1998**, p. 421.

[45] a) G. K. S. Prakash in *Synthetic Fluorine Chemistry* (Eds.: G. A. Olah, R. D. Chambers, G. K. S. Prakash), Wiley, New York, **1992**, p. 227; b) G. K. S. Prakash, A. K. Yudin, *Chem. Rev.* **1997**, *97*, 757; c) M. Fujita, M. Obayashi, T. Hiyama, *Tetrahedron* **1988**, *44*, 4135.

[46] For reviews on oxiranyl anions, see: a) T. Satoh, *Chem. Rev.* **1996**, *96*, 3303; b) Y. Mori, *Rev. Heteroat. Chem.* **1997**, *17*, 183; c) D. M. Hodgson, E. Gras, *Synthesis* **2002**, 1625; see also: d) G. A. Molander, K. Mautner, *Pure Appl. Chem.* **1990**, *62*, 707; e) E. Doris, L. Dechoux, C. Mioskowski, *Synlett* **1998**, 337.

[47] a) M. Shimizu, T. Fujimoto, H. Minezaki, T. Hata, T. Hiyama, *J. Am. Chem. Soc.* **2001**, *123*, 6947; b) M. Shimizu, T. Fujimoto, X. Liu, H. Minezaki, T. Hata, T. Hiyama, *Tetrahedron* **2003**, *59*, 9811.

[48] It is widely known that β-oxido carbenoids undergo 1,2-rearrangement of a substituent β to the oxygen atom, giving rise to carbonyl compounds; for reviews, see: a) P. M. Wockulich, in *Comprehensive Organic Synthesis*, *Vol. 1* (Eds.: B. M. Trost, I. Fleming), Pergamon, Oxford, **1991**, p. 873; b) G. Boche, J. C. W. Lohrenz, *Chem. Rev.* **2001**, *101*, 697; for examples, see: c) J. Villieras, C. Bacquet, J. F. Normant, *J. Organomet. Chem.* **1972**, *40*, C1; d) G. Köbrich, J. Grosser, *Tetrahedron Lett.* **1972**, 4117; e) H. Taguchi, H. Yamamoto, H. Nozaki, *Tetrahedron Lett.* **1972**, 4661; f) G. Köbrich, J. Grosser, *Chem. Ber.* **1973**, *106*, 2626; g) J. Villieras, C. Bacquet, J. F. Normant, *J. Organomet. Chem.* **1975**, *97*, 325; h) H. Taguchi, H. Yamamoto, H. Nozaki, *Bull. Chem. Soc. Jpn.* **1977**, *50*, 1592.

[49] Y. Yamauchi, T. Katagiri, K. Uneyama, *Org. Lett.* **2002**, *4*, 173.

[50] For reviews on oxepanes and oxepins, see: a) D. R. Boyd in *Comprehensive Heterocyclic Chemistry*, *Vol. 7* (Eds.: A. R. Katritzky, C. W. Rees), Pergamon, Oxford, **1984**, p. 547; b) W. Tochtermann, G. Olsson, *Chem. Rev.* **1989**, *89*, 1203; c) L. I. Belen'kii in *Comprehensive Heterocyclic Chemistry II*, *Vol. 9* (Eds.: A. R. Katritzky, C. W. Rees, E. F. V. Scriven), Pergamon, Oxford, **1996**, p. 45; d) D. R. Boyd, N. D. Sharma, *Chem. Soc. Rev.* **1996**, *25*, 289; e) S. von Angerer, W. Tochtermann in *Methods of Organic Chemistry*, *Vol. E9 (Houben-Weyl)*, Thieme, Stuttgart, **1998**, p. 1.

[51] For a review on the Cope rearrangement of bis(alkenyl)oxiranes, see: a) T. Hudlicky, R. Fan, J. W. Reed, K. G. Gadamasetti, *Organic Reactions*, *Vol. 41*, Wiley, New York, **1992**, p. 1; for examples, see: b) T. Hudlicky, A. Fleming, T. C. Lovelace, *Tetrahedron* **1989**, *45*, 3021; c) D. L. Clark, W.-N. Chou, J. B. White, *J. Org. Chem.* **1990**, *55*, 3975; d) W.-N. Chou, J. B. White, *Tetrahedron Lett.* **1991**, *32*, 157; e) W.-N. Chou, J. B. White, W. B. Smith, *J. Am. Chem. Soc.* **1992**, *114*, 4658; f) P. von Zezschwitz, K. Voigt, A. Lansky, M. Noltemeyer, A. de Meijere, *J. Org. Chem.* **1999**, *64*, 3806.

[52] M. Shimizu, T. Fujimoto, X. Liu, T. Hiyama, *Chem. Lett.* **2004**, *33*, 438.

[53] *Preparation of Alkenes: A Practical Approach* (Ed.: J. M. J. Williams), Oxford University Press, Oxford, **1996**, p. 1.

[54] a) *Drugs Fut.* **1985**, *10*, 395; b) *Drugs Fut.* **1990**, *15*, 532; c) G. Abraham, T. Horvath, L. Toldy, J. Borvendeg, E. Csanyi, E. Kiss, I. Hermann, S. K. Tory (Gyogyszerkutato Intezet), US 763078, **1989** [*Chem. Abstr.* **1989**, *111*, 77630]; for biological properties, see: d) J. Borvendeg, I. Hermann, O. Csuka, *Acta Physiol. Hung.* **1996**, *84*, 405 [*PubMed ID*, **1996**, 9328614]; e) V. Erdelyi-Toth, F. Gyergyay, I. Szamel, E. Pap, J. Kralovanszky, E. Bojti, M. Csorgo, S. Drabant, I. Klebovich, *Anti-Cancer Drugs* **1997**, *8*, 603 [*Chem. Abstr.* **1997**, *127*, 287616]; for the synthesis of panomifene, see: f) G. Nemeth, R. Kapiller-Dezsofi, G. Lax, G. Simig, *Tetrahedron* **1996**, *52*, 12821.

[55] a) Y. Ukaji, T. Fujisawa, *Tetrahedron Lett.* **1988**, *29*, 5165; b) Y. Ukaji, A. Yoshida, T. Fujisawa, *Chem. Lett.* **1990**, 157; c) M. Taniguchi, K. Oshima, K. Utimoto, *Tetrahedron Lett.* **1991**, *32*, 2783; d) B. Santiago, C. Lopez, J. A. Soderquist, *Tetrahedron Lett.* **1991**, *32*, 3457; e) J. A. Soderquist, C. Lopez, *Tetrahedron Lett.* **1991**, *32*, 6305; f) E. Doris, L. Dechoux, C. Mioskowski, *Tetrahedron Lett.* **1994**, *35*, 7943; g) E. Doris, L. Dechoux, C. Mioskowski, *Tetrahedron Lett.* **1994**, *35*, 7943; h) A. N. Kasatkin, R. J. Whitby, *Tetrahedron Lett.* **2000**, *41*, 5275.

[56] a) T. Hata, H. Kitagawa, H. Masai, T. Kurahashi, M. Shimizu, T. Hiyama, *Angew. Chem.* **2001**, *113*, 812; *Angew. Chem. Int. Ed.* **2001**, *40*, 790; b) M. Shimizu, H. Kitagawa, T. Kurahashi, T. Hiyama, *Angew. Chem.* **2001**, *113*, 4283; *Angew. Chem. Int. Ed.* **2001**, *40*, 4413; c) M. Shimizu, T. Kurahashi, T. Hiyama, *Synlett* **2001**, 1006; d) T. Kurahashi, T. Hata, H. Masai, H. Kitagawa, M. Shimizu, T. Hiyama, *Tetrahedron* **2002**, *58*, 6381; e) M. Shimizu, T. Kurahashi, H. Kitagawa, T. Hiyama, *Org. Lett.* **2002**, *4*, 225; f) M. Shimizu, T. Kurahashi, H. Kitagawa, K. Shimono, T. Hiyama, *J. Organomet. Chem.* **2003**, *686*, 286.

[57] a) M. W. Rathke, R. Kow, *J. Am. Chem. Soc.* **1972**, *94*, 6854; b) D. S. Matteson, *Synthesis* **1975**, 147.

[58] Triaryl alkenes are one of the key structures of antiestrogens; for reviews on antiestrogens, see: a) R. A. Magarian, L. B. Overacre, S. Singh, K. L. Meyer, *Curr. Med. Chem.* **1994**, *1*, 61; b) V. C. Jordan, *J. Med. Chem.* **2003**, *46*, 883; c) V. C. Jordan, *J. Med. Chem.* **2003**, *46*, 1081.

[59] X. Liu, M. Shimizu, T. Hiyama, *Angew. Chem.* **2004**, *115*, 897; *Angew. Chem. Int. Ed.* **2004**, *43*, 879.

[60] a) *Asymmetric Fluoroorganic Chemistry: Synthesis, Applications, and Future Directions*, *ACS Symposium Series 746* (Ed.: P. V. Ramachandran), American Chemical Society, Washington, **2000**; b) *Enantiocontrolled Synthesis of Fluoro-Organic Compounds* (Ed.: V. A. Soloshonok), Wiley, Chichester, **1999**; c) K. Iseki, *Tetrahedron* **1998**, *54*, 13887; d) T. Kitazume, T. Yamazaki, *Top. Curr. Chem.* **1997**, *193*, 90; e) P. Bravo, G. Resnati, *Tetrahedron: Asymmetry* **1990**, *1*, 661.

[61] T. Hiyama, T. Kusumoto, H. Matsutani in *Asymmetric Fluoroorganic Chemistry: Synthesis, Applications, and Future Directions*, *ACS Symposium Series 746* (Ed.: P. V. Ramachandran), American Chemical Society, Washington, **2000**, p. 226.

[62] H. Poras, H. Matsutani, J. Yaruva, T. Kusumoto, T. Hiyama, *Chem. Lett.* **1998**, 665.

[63] H. Matsutani, H. Poras, T. Kusumoto, T. Hiyama, *Chem. Commun.* **1998**, 1259.

[64] H. Matsutani, S. Ichikawa, J. Yaruva, T. Kusumoto, T. Hiyama, *J. Am. Chem. Soc.* **1997**, *119*, 4541.
[65] a) T. Katagiri, H. Ihara, M. Takahashi, S. Kashino, K. Furuhashi, K. Uneyama, *Tetrahedron: Asymmetry* **1997**, *8*, 2933; b) T. Katagiri, M. Irie, K. Uneyama, *Tetrahedron: Asymmetry* **1999**, *10*, 2583.
[66] H. Matsutani, H. Poras, T. Kusumoto, T. Hiyama, *Synlett* **1998**, 1353.
[67] H. Matsutani, T. Kusumoto, T. Hiyama, *Chem. Lett.* **1999**, 529.